

Topic 1: Measurement and Uncertainties

1.1

Deriving Fundamental Units

Ex. What are the fundamental SI units of power?

Step 1: Break Down Eq

$$P = \frac{W}{t} \quad P = \frac{Fd}{t} \quad P = \frac{\vec{m}ad}{t}$$

Step 2: Sub in SI units

$$P = \frac{(kg)(m/s^2)(m)}{s}$$

Step 3: Simplify:

$$P = \frac{kg \, m^2}{s^3}$$

Significant Digits

When given a question, locate the value with the lowest number of significant figures and limit your answer accordingly.

1.2

Propagating Error

$$y = a \pm b \quad \Delta y = \Delta a + \Delta b$$

When adding or subtracting in the same units of uncertainty add/subtract the errors to the least known digit.

$$y = \frac{ab}{c} \quad \Delta y = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c}$$

When multiplying or dividing the net uncertainty is the sum of all relative errors (percent form)

$$y = a^n \quad \Delta y = \left| n \frac{\Delta a}{a} \right|$$

When propagating an exponent/radical the uncertainty is multiplied by the exponent, note: put radical into exponent form.

Topic 2: Mechanics

2.1: Motion

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad v^2 = u^2 + 2as \quad s = \frac{(v+u)t}{2}$$

Displacement and Distance

Displacement: The final distance from endpoint relative to the original starting point. (vector)

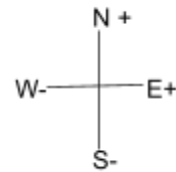
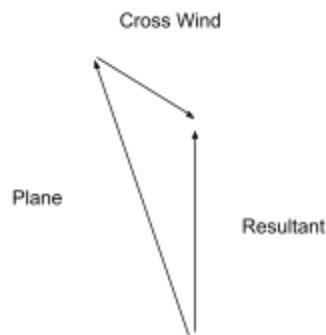
Distance: The total length traveled in any direction. (scalar)

Speed and Velocity

$$v = \frac{s}{t} \quad \vec{v} = \frac{\Delta \vec{s}}{\Delta t}$$

Vectors and Relative Motion

A plane wishes to fly from Calgary to Edmonton which is 298 km due North of Calgary. The wind is reported to be blowing at 75.4 km/h [21.3 S of E]. If the plane can fly at 640 km/h in non windy conditions. What direction must the plane aim to fly directly to Edmonton? How much time is required.



$$s_{\text{edmonton}} = 298 \text{ km [N]} \quad v_{\text{wind}} = 75.4 \text{ km/h [21.3}^\circ \text{ S of E]} \quad v_R = ? \text{ [N]} \quad v_{\text{plane}} = 640 \text{ km/h}$$

$$\sum v_x = 0 \quad 0 = \cos(21.3)(75.4 \text{ km/h}) + v_{x\text{-plane}} \quad v_{x\text{-plane}} = \cos(21.3)(75.4 \text{ km/h}) \text{ [W]}$$

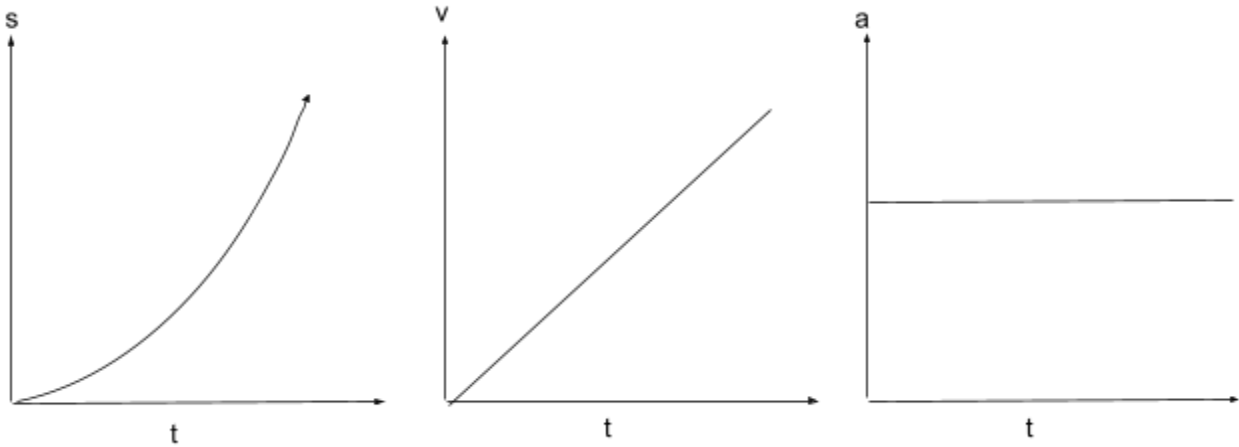
$$\theta = \cos^{-1}\left(\frac{\cos(21.3)(75.4 \text{ km/h})}{640 \text{ km/h}}\right) = 83.7^\circ \text{ [N of W]}$$

$$v_p^2 = v_{xp}^2 + v_{yp}^2$$

Acceleration

$$a = \frac{v-u}{\Delta t}$$

Graphical Interpretation



Velocity is the slope of the position time graph.

Acceleration is the slope of the velocity time graph.

Projectile Motion

A soccer ball is kicked at 45m/s 30 degrees above the horizontal, what is the range and max height?

Knowns: $\vec{a}_y = -9.81 \text{ m/s}^2$ $v_{y-top} = 0$ $u_y = 45 \text{ m/s } [30^\circ]$

$$u_y = \sin(30)(45 \text{ m/s}) \quad u_x = \cos(30)(45 \text{ m/s})$$

Key: Time is what connects projectile questions

$$v = u + at$$

$$t = \frac{v_y - u_y}{a_y}$$

$$t = \frac{0 \text{ m/s} - (\sin(30)(45 \text{ m/s}))}{-9.81}$$

$$t = 4.5 \text{ s}$$

From here max height can be found either using $v^2 = u^2 + 2as$ or $s = ut + \frac{1}{2}at^2$

To find max length

$$(t \times 2)(\cos(30))(45m/s) = s$$

Fluid Resistance and Terminal Speed

As an object accelerates in free fall the force of air resistance will eventually be equal to the force of gravity. When these two forces are equal the object is at terminal velocity and is at constant speed.

2.2 Forces

Newton's Law

1st Law

Every object in motion or at rest, stays in motion or at rest unless externally acted upon. Every object's tendency to remain in the state of motion is called **Inertia**.

$$F_{net} = 0 \text{ at constant velocity or rest}$$

2nd Law

Any object of mass acted on by an external force will accelerate

$$F = ma$$

3rd Law

Every force has an equal and opposite force as a result of it.

Types of forces

Force of gravity is an object's mass multiplied by the gravitational constant. The force of gravity acts on all objects.

$$F_g = mg$$

The **force of friction** is the normal force multiplied by the static/kinetic frictional constant.

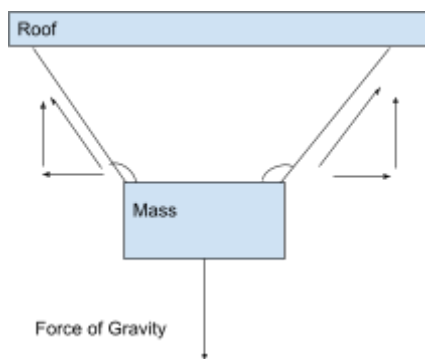
$$F_f = \mu F_N$$

The **force normal** is the force acting in opposition to gravity and other forces. Usually it is the equal opposite force of gravity, but other forces such as tension, angle of plane can affect it.

Note: Forces are vectors and can be broken into components just like velocity etc.

Standard Dynamic Problems

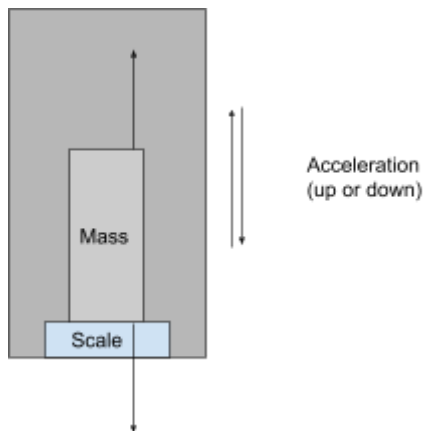
Translational Equilibrium



Key Connections

$$\vec{F}_{net} = 0 \quad \vec{F}_g = \Sigma \vec{F}_{Ty}$$

Elevator



Key Connections

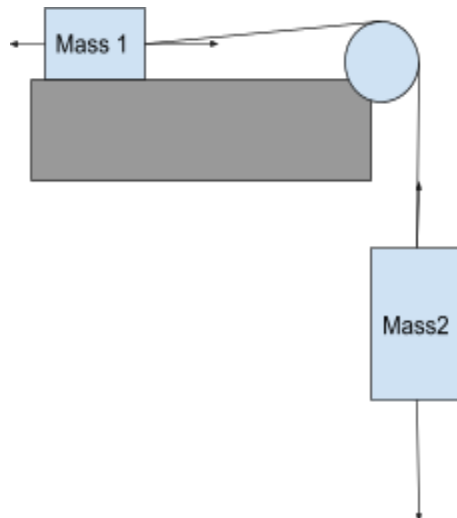
If acceleration of elevator is 0 (constant speed / rest)

$$F_g = F_N$$

If acceleration present

$$\vec{F}_{net} = m\vec{a} = \vec{F}_g + \vec{F}_N$$

Pulley



Key Connections

Treat apparatus as one object where tension does not exist.

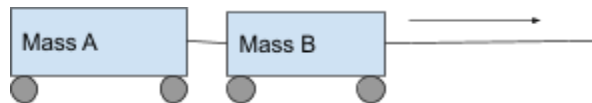
$$\vec{F}_{net} = m_{1+2}\vec{a} = \vec{F}_g + \vec{F}_f$$

To find tension separate into individual objects

$$\vec{F}_{net\ 1} = \vec{F}_f + \vec{F}_T = m_1\vec{a}$$

Note: Acceleration is the same for both the whole apparatus and individual objects

Train



Key connections

Calculate the acceleration of the whole train with Newton's Second law. $\vec{F}_{net} = m_{A+B}\vec{a}$
Using the acceleration other forces can then be found by looking at each object individually

$$\vec{F}_{net\ B} = \vec{F}_{AB} + \vec{F}_T = m_B\vec{a}$$

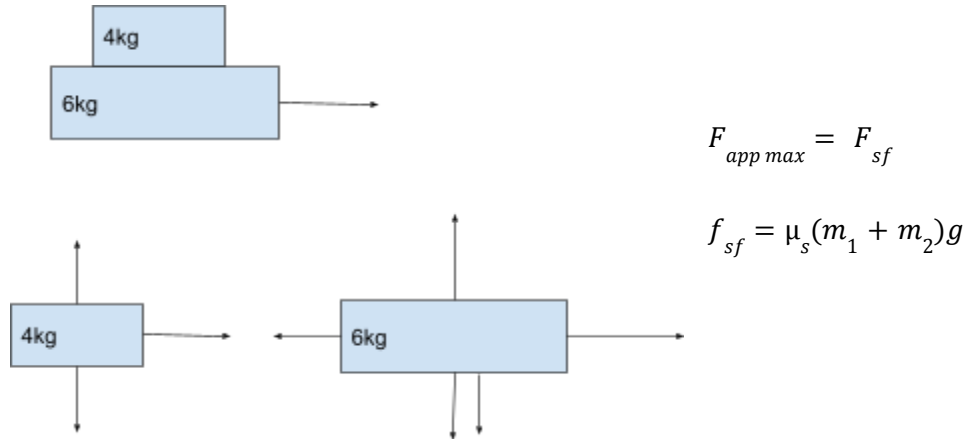
Note: Acceleration is the same for both the whole apparatus and individual objects

Boxes

Connection: Acceleration of the whole apparatus is the same for each individual box.

Hard IB Mechanic Questions

A block of mass 4.0 kg is placed on top of a block of mass 6.0 kg. The static coefficient of friction between the two blocks is 0.60 and the kinetic coefficient is 0.50. What is the largest force that can be applied to the 6.0 kg block so that the 4.0 kg block does not slip?



2.3 Work, Energy and Power

Work

When a force acts on an object over a displacement. Measure in Joules.

$$W = F\Delta s \cos\theta$$

If the force is not completely in the direction of displacement the component of the force that is parallel to the displacement must be used.

Note: If a force is in the opposite direction of displacement it is negative.

Net Work

$$W_{net} = W_1 + W_2 \dots \quad W_{net} = \Delta E_k$$

Kinetic Energy

All objects in motion have kinetic energy.

$$E_k = \frac{1}{2}mv^2$$

Gravitational Potential Energy

$$E_{pg} = mg\Delta h$$

Elastic Potential Energy

$$E_{ps} = \frac{1}{2}kx^2$$

Conservation of Energy

$$E_{mi} = E_{mf} \quad \text{when energy is fully conserved}$$

$$E_{mi} + W_{NC} = E_{mf}$$

Power

$$P = \frac{E}{t}$$

Efficiency

$$\text{efficiency} = \frac{\text{Useful Power Out}}{\text{Useful Power In}}$$

2.4

Impulse

$$\text{Impulse} = F\Delta t \quad F\Delta t = \Delta p$$

A 500g ball is dropped from a height of 4.0 meters and falls toward the ground.

a) Determine the speed just before it hits the ground.

b) The ball bounces and rises up to a height of 3.0m, estimate the average force from the floor on the ball if the floor is in contact with the ball for 6.0 milliseconds. (Answer: 1377 N upward)

$$mgh = \frac{1}{2}mv^2$$

$$(9.81)(4) = \frac{1}{2}v^2$$

$$v = \sqrt{(9.81)(4)(2)}$$

$$v = 8.85 \text{ m/s [Down]}$$

$$\frac{1}{2}mv^2 = mgh$$

$$v = \sqrt{(9.81)(3)(2)}$$

$$v = 7.67 \text{ [Up]}$$

$$F\Delta t = \Delta p$$

$$F\Delta t = mu - mv$$

$$F = \frac{(0.5\text{kg})(7.67\text{m/s}) - (0.5\text{kg})(-8.85\text{m/s})}{0.006\text{s}}$$

$$F = 1377 \text{ N [up]}$$

Conservation of linear momentum

$$\sum p = \sum p' \quad p = mv$$

Hit and Bounce

Hit and Stick

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad m_1 u_1 + m_2 u_2 = (m_1 + m_2) v_{1+2}$$

Explosion

Important: Create Conventions and express direction with sign

$$(m_1 + m_2) u_{1+2} = m_1 v_1 + m_2 v_2$$

Elastic Collisions: Kinetic energy conserved through collision. $E_{kf} - E_{ki} = 0$

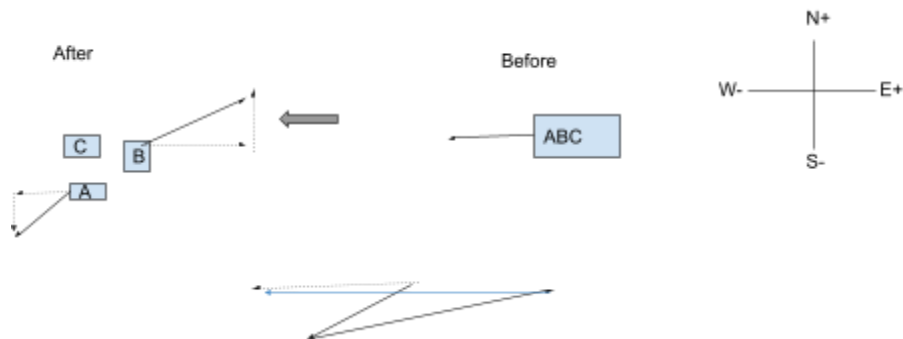
Inelastic Collision: Kinetic energy is lost from collision. $E_{kf} - E_{ki} \neq 0$

2D Momentum

$$\sum p_x = \sum p'_x \quad \sum p_y = \sum p'_y$$

When tackling 2D momentum problems remember both components of momentum are conserved.

An object with momentum 30 kg m/s West explodes and three pieces fly off. One piece has momentum 75 kgm/s 20°N of E, a second piece with momentum of 50 kg m/s at 45° S of W. What is the magnitude of momentum of the third piece?



$$\sum p_x = \sum p'_x \quad (-30 \text{ kgm/s}) = \cos(20)(75 \text{ kgm/s}) + (-\cos(45)(50 \text{ kgm/s})) + p_{Cx}$$

$$\sum p_y = \sum p'_y \quad 0 = \sin(20)(75 \text{ kgm/s}) + (-\sin(45)(50 \text{ kgm/s})) + p_{Cy}$$

$$p_C = \sqrt{(p_{Cx})^2 + (p_{Cy})^2} \quad p_C = \sqrt{(-65.12 \text{ kgm/s})^2 + (9.703 \text{ kgm/s})^2}$$

$$\theta = \tan^{-1}(9.703/65.12) \quad p_C = 65 \text{ kgm/s } [8.5^\circ \text{ N of W}]$$

Topic 3: Thermal Physics

3.1 Thermal Concepts

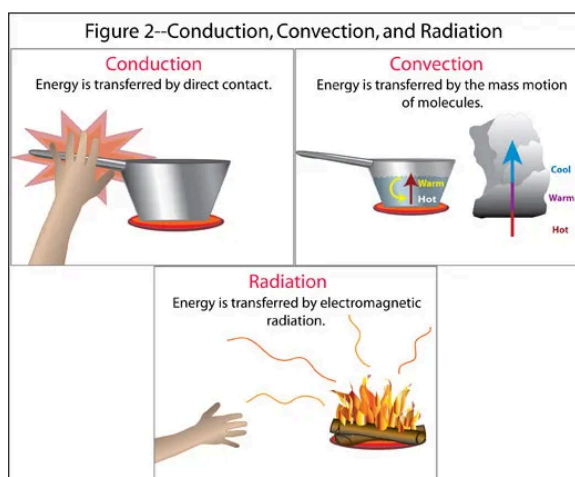
Molecular theory of solids, liquids and gasses

Solids: The molecules that make up a solid are arranged in regular repeating patterns. Low kinetic energy, particles are held firm but can vibrate.

Liquid: Liquids possess more kinetic energy than solids and are held together by intermolecular forces. Because there is enough kinetic energy the particles are free to flow and assume shapes of containers.

Gas: Gasses have high kinetic energy which creates less opportunities for molecules to interact and create intermolecular forces. The high kinetic energy makes particles fly and shoot everywhere.

Plasma: At very high temperature atoms start to lose their electrons. The mixture of electrons and nuclei that result is the plasma.



Temperature and absolute temperature

Temperature is the measure of kinetic energy within a substance, temperature can describe how hot or cold an object is and determines the direction of heat flow. Thermal energy transfers from an object with higher temperature to another object with lower temperature. This transfer is called **heat**.

Absolute zero: lowest possible temperature (0K -273.15 c)

Internal Energy

The internal energy of a substance is the sum of kinetic energy (thermal energy) and total potential energy (energy stored in bonds).

Specific Heat Capacity

A substance's heat capacity dictates the heat required to change the temperature of a substance.

$$Q = mc\Delta T \quad Q = \text{Heat (J)} \quad m = \text{mass (kg)} \quad c = \text{specific heat capacity} \quad \Delta T = \text{change in temp}$$

Phase Change

Change of Phase	Process	Kinetic Energy	Potential Energy
Solid to Liquid	Melting	Unchanged	Increase
Liquid to Solid	Freezing	Unchanged	Decrease
Liquid to Gas	Boiling	Unchanged	Increase
Gas to Liquid	Condensation	Unchanged	Decreases

During a phase change, temperature and kinetic energy remain constant and potential energy changes (which increases as molecules spread out and vice versa)

Specific Latent Heat

When an object reaches a temperature at which it changes phase, it maintains that temperature until the phase change is complete. **Specific Latent Heat** is the amount of heat required for 1 kg of a substance to undergo phase change.

$$Q = mL \quad Q = \text{Heat Change} \quad m = \text{mass} \quad L = \text{Specific Latent Heat}$$

3.2

Pressure

$$P = \frac{\text{Force}}{\text{Area}}$$

Equation of state for an ideal gas

$$PV = nRT$$

$$P = \text{Pressure (kPa)} \quad V = \text{Volume (L)} \quad n = \text{mole (mol)} \quad R = \text{ideal gas constant} \quad T = \text{Temperature (K)}$$

Kinetic model of an ideal gas

When looking at the model of an ideal gas there are many assumptions made.

- Collisions between molecules are perfectly elastic
- The molecules are identical spheres
- The volume of molecules is negligible compared to the volume of the gas
- Intermolecular forces do not exist
- Under high temperature, low pressure/low density a real gas behaves like a ideal gas.

Mole, molar mass and the Avogadro Constant

Avogadro's constant states that for every mol of a substance there are 6.022×10^{23} constituent particles within.

Topic 4: Waves

4.1 Oscillations

Simple Harmonic Oscillation

When an object moves about a fixed position back and forth with a periodic pattern. For simple harmonic oscillation a restoring force must be acting on the object and no loss of energy can occur.

Common oscillators: The simple pendulum and a harmonic spring oscillator are two very common examples of harmonic motion

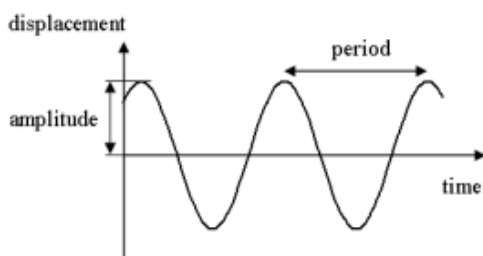
$$T = 2\pi\sqrt{\frac{l}{g}} \quad T = 2\pi\sqrt{\frac{m}{k}}$$

$T = \text{period (s)}$ $l = \text{length (m)}$ $g = 9.81 \text{ m/s}^2$ $m = \text{mass (kg)}$ $k = \text{spring constant (N/m)}$

Damped Harmonic Oscillator

Harmonic oscillators in the real world can never really have perfect harmonic motion due to external forces (friction, air resistance etc). Damped harmonic oscillators will eventually come to rest.

Time period, frequency, amplitude, displacement and phase difference



Period: The time for one oscillation

Frequency: The amount of an oscillation per second.

$$T = \frac{1}{f}$$

Amplitude: The maximum displacement

Displacement: An object's position relative to equilibrium at a given time.

Phase Difference: The difference between two SMH oscillators with the same frequency in terms of their relative position in cycle measured in radians.

Conditions for simple harmonic motion

For an object to be in constant harmonic motion, there must be a restoring force constantly directed towards the equilibrium point.

The force must always be proportional to the displacement of the object. As the further an object is from equilibrium the harder the restoring force acts.

4.2 Travelling Waves

Traveling Waves

A traveling wave is the continuous disturbance of motion in the form of oscillations within a medium. A wave is the transfer of energy, and the direction of the wave is determined by the direction of energy transfer.

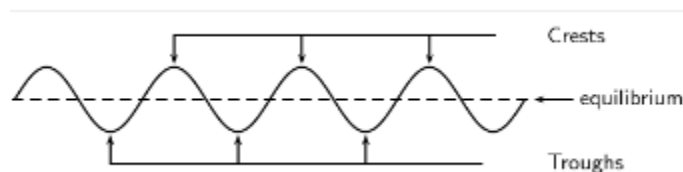
Wavelength, frequency, period and wave speed

$$v = f\lambda \quad v = \text{speed (m/s)} \quad f = \text{frequency (Hz)} \quad \lambda = \text{wavelength (m)}$$

The speed of a wave is determined by the frequency and wavelength of a wave.

Transverse Waves: A transverse wave has oscillations perpendicular to the displacement of the wave. Ex. Rope

Characteristics: A transverse wave has crests at highest displacement and troughs at lowest displacement, the wavelength of a wave would be the distance between two points of exact location on 2 waves (crest to crest). The location at which displacement is zero is called equilibrium.

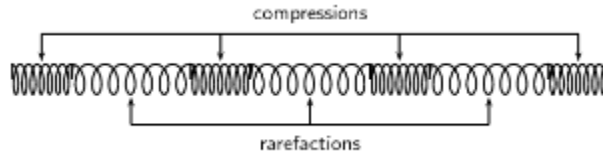


You are standing on a dock watching a nearby boat moving up and down on the waves. The time it takes for the boat to move from equilibrium to a crest is 0.84s. The horizontal distance from a crest to the 2nd trough away from it is 12.9 m. What is the speed of the wave?

$$T = 0.84s \times 4 \quad \lambda = 2 \times 3(12.9)m \quad v = f\lambda = \frac{\lambda}{T} = 2.56 m/s$$

Longitudinal Wave: A longitudinal wave has oscillations in the same plane of displacement.
Ex. sound.

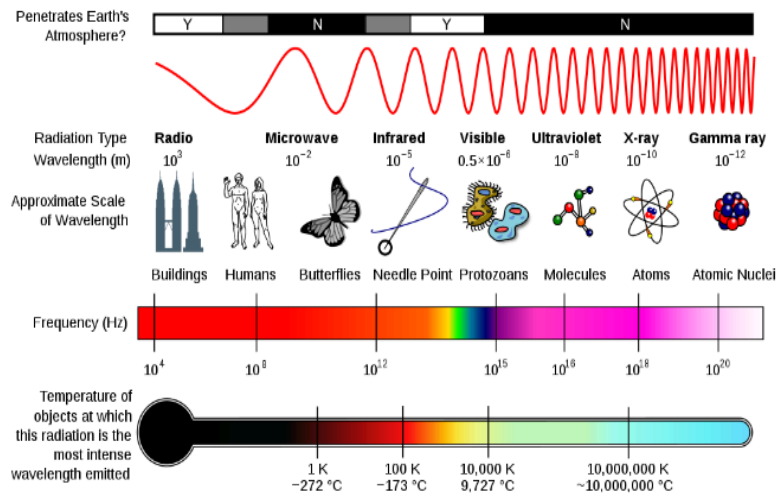
Characteristics: A longitudinal wave has points of **compression** and **rarefaction** at which particles are either clumped the most or spread the most.



The nature of electromagnetic waves

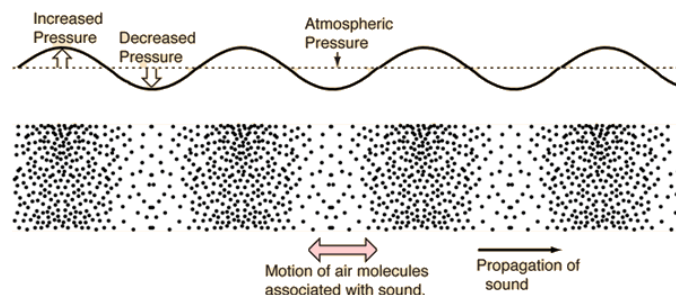
All electromagnetic waves have the speed of light, the frequency or wavelength determines the type of electromagnetic radiation it possesses. Electromagnetic waves of higher frequency contain higher energy.

$$\lambda \propto \frac{1}{f}$$



The nature of sound waves

The speed of sound at 20 degrees celsius is around 343.2m/s. All sound waves are longitudinal



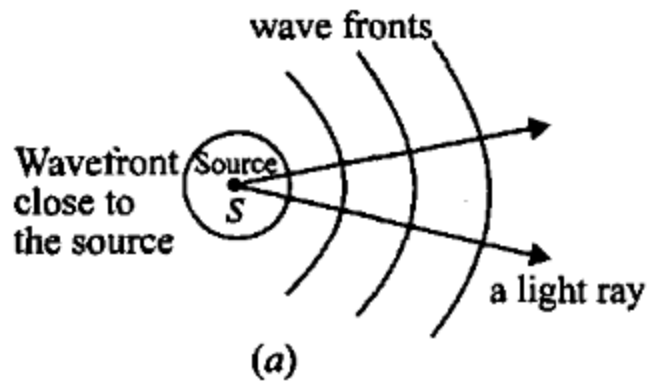
4.3 Wave Characteristics

Wavefronts

Wavefronts are the lines joining all vibrations in phase, the distance from wavefront to wavefront is the wavelength.

Rays

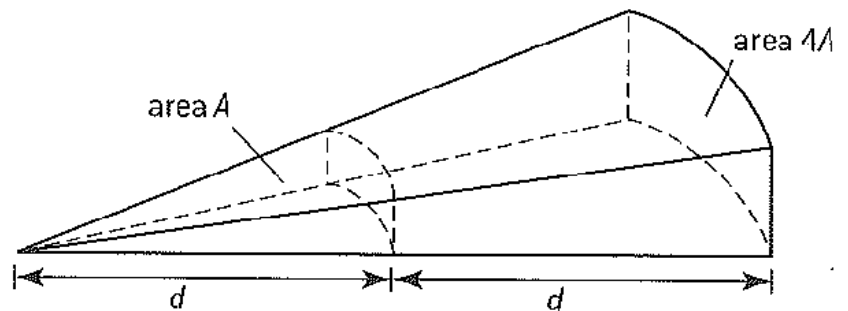
Rays show the direction of wave propagation, rays are always perpendicular to the wavefronts.



Amplitude and intensity

Amplitude the displacement from the equilibrium point of a wave to its crest or trough. Intensity is the power delivered per unit area. The intensity of a wave is proportional to the square of amplitude.

$$I \propto A^2$$

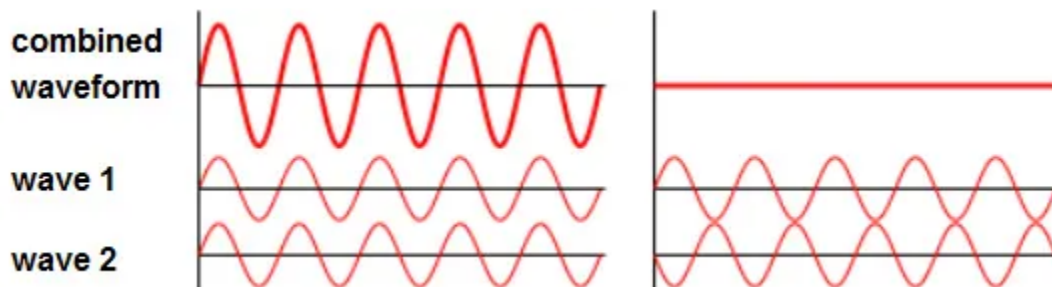


Also in relation to a wave, the Intensity of wave is proportional to the inverse square of the distance from source.

$$I \propto \frac{1}{x^2} \quad \text{because of the proportionality} \quad \frac{I_1}{I_2} = \left(\frac{x_2}{x_1} \right)^2$$

Superposition

The principle of superposition states that the net displacement of a medium undergoing multiple oscillations will be the sum of each individual displacement.



Polarization

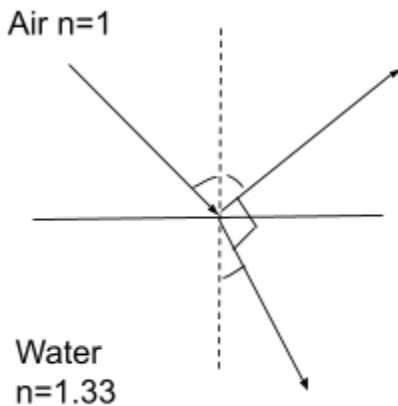
Polarized light is light in which the transverse wave occurs only on a single plane. Unpolarized light contains infinitely large amounts of transverse waves oscillating on all planes.

Polarization from reflection

When the angle of refraction and reflection form a 90 degrees angle light is both reflected and refracted. The reflected light will be linearly polarized. The angle at which light is polarized is called the Brewster's angle.

$$\tan(\theta) = \frac{n_2}{n_1} \quad n_1 = \text{incident index} \quad n_2 = \text{refractive index} \quad \theta = \text{Brewster's angle}$$

Light travels from air ($n=1$) to water ($n=1.33$). At what incident angle is the reflected light completely polarized? What is the angle of refraction for which light is transmitted to the water.



$$\tan(\theta) = \frac{n_2}{n_1} \quad \theta = \tan^{-1}\left(\frac{1.33}{1}\right) \quad \theta = 53.06$$

$$\frac{n_1}{n_2} = \frac{\sin\theta_2}{\sin\theta_1}$$

$$\theta_2 = \sin^{-1}\left(\frac{\sin(53.06)(1)}{1.33}\right)$$

$$\theta_2 = 36.9$$

Polarizers and Analyzers

Malus' law relates the incident intensity and transmitted intensity of light passing through a polarizer and an analyzer.

When unpolarized light passes through a polarizer the intensity of light is reduced by 50%. Once partially polarized, if the light were to undergo another polarizer, the angle of rotation would determine the effectiveness of the second polarizer. A polarizer that can be orientated is called an analyzer.

$$I = I_0 \cos^2(\theta)$$

I_0 = Intensity after passing through a polarizer (reduced 50%)

I = Intensity after passing through both a polarizer and analyzer (reduced > 50%)

θ = Angle at which analyzer is orientated

4.4

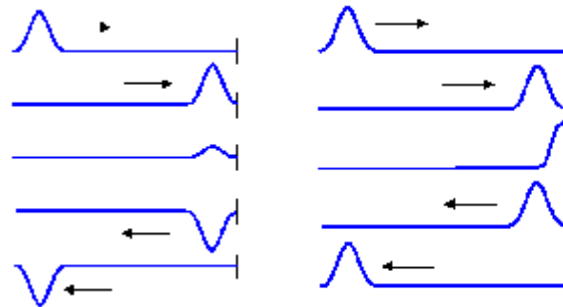
Refraction and Reflection

Reflection is when a wave reflects off of a medium. Refraction is when light passes through a medium.

Behavior

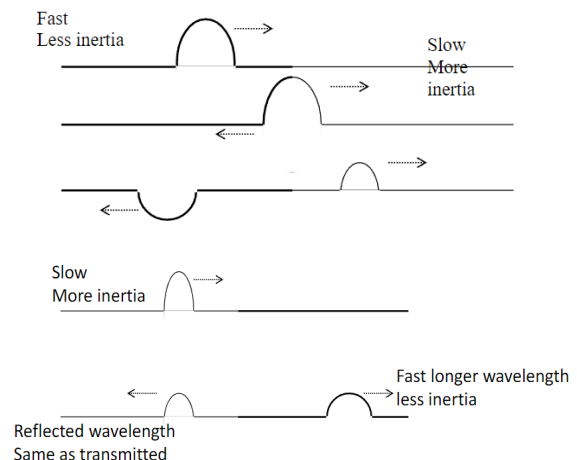
When a transverse wave is oscillated with one end fixed, the wave is reflected back displaced oppositely.

When a wave hits an open end it reflects exactly the same as the incident properties.



Medium Change

When a wave enters a medium that is more dense or heavier the more stiff medium acts like a fixed end and the wave is both reflected back and slowed down. When a wave enters a medium that is less stiff the lighter medium acts like a free end and the wave is reflected on the same side and sped up.



Snell's Law, Critical Angle, Total Internal Inflection

Snell's law relates the angle of incidence and angle of refraction by their individual refractive indexes.

$$\frac{n_2}{n_1} = \frac{\sin \theta_1}{\sin \theta_2} \quad n_1, \theta_1 = \text{incident index and angle} \quad n_2, \theta_2 = \text{refracted index and angle}$$

Critical Angle: Is the angle at which light no longer refracts anymore but refracts light along the surface and reflects light. To find the critical angle of a medium change set the refracted angle to 90

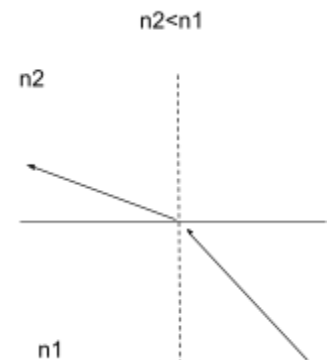
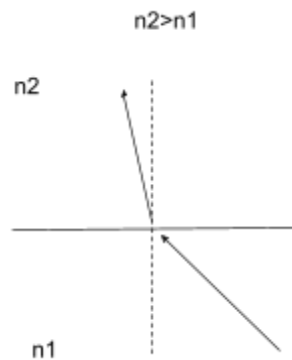
$$\frac{n_2}{n_1} = \frac{\sin \theta_1}{\sin \theta_2} \quad \text{when refracted angle is } 90 \quad \frac{n_2}{n_1} = \frac{\sin \theta_{\text{critical}}}{\sin (90)}$$

Total internal inflection: Any light past the critical angle is completely reflected off the surface. Snell law error.

$$\text{Angle of Incidence} = \text{Angle of Reflection}$$

Medium Change Behaviors

When an object goes from a less dense medium to a more dense medium the wavelength is shortened and speed is reduced. This causes the wave to refract towards the normal and vice versa. Key: Frequency is always constant.



Because frequency is always the

same. $\frac{v_1}{\lambda_1} = \frac{v_2}{\lambda_2}$

Diffraction

When even wavefronts hit a small opening or object the wavefronts are diffracted and sent radially.

Interference Patterns

When two waves with no phase change interact with each other and their respective antinodes interfere to change net displacement. Waves can interfere either due to path difference (length of waves are different) or phase difference (antinodes created at different moments).

Constructive Interference: When the antinodes of both waves line up (both displaced upwards/downwards) and constructively add up.

Destructive Interference: When the antinodes of two waves line up but are displaced oppositely, destructively interfering and reducing net displacement.

Double Slit Interference (due to path difference)

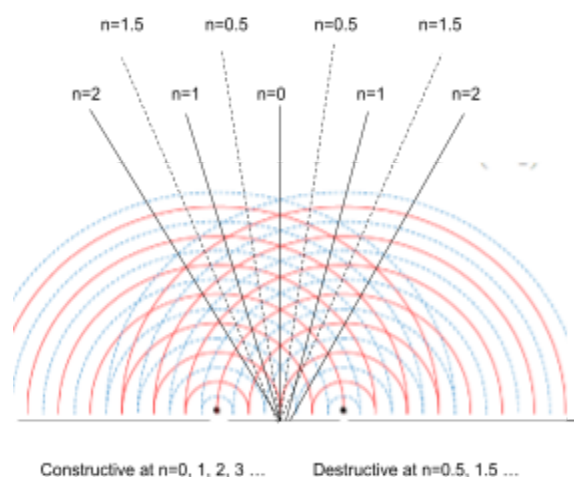
Due to diffraction when wavelengths hit corners or obstacles, waves bend and refract. The double slit is when waves get diffracted by two interferences and create nodal lines in which light can be more defined (due to constructive interference).

$$s = \frac{\lambda D}{d}$$

s = distance between fringe (0 - 1)

λ = wavelength

D = distance to wall



d = distance between slits

Not Given by IB

$$\lambda = \frac{d \sin \theta}{n} \text{ and } \tan \theta = s/D$$

θ = angle between normal and nodal line

The distance from slit plates to screen is 4.5 meters the distance from central dot to the second dot or maxima is 1.2cm

•A) What is the angle between the 2nd maxima and central bright spot

$$\tan \theta = s/D \quad \tan \theta = (0.012\text{m})/4.5\text{m}$$

$$\theta = \tan^{-1}(0.012\text{m}/4.5\text{m}) \quad \theta = 0.15^\circ$$

•B) if light of wavelength 726 nm is shone through the double slit, what is the distance between the openings on the slit plate?

$$s = \frac{\lambda D}{d}$$

$$d = \frac{\lambda D}{s}$$

$$d = \frac{(726 \times 10^{-9}\text{m})(4.5\text{m})}{(0.006\text{m})}$$

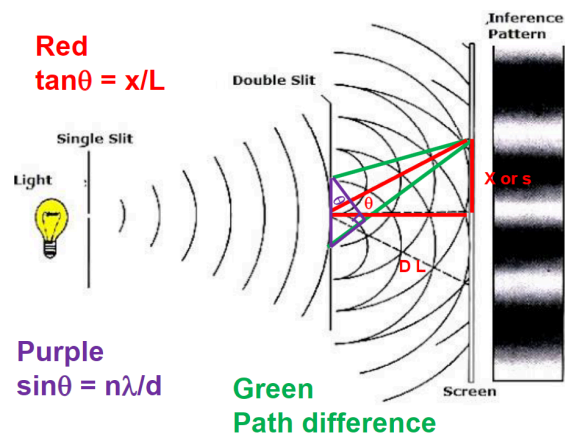
$$d = 5.45 \times 10^{-4}\text{m}$$

$$\text{or} \quad \lambda = \frac{d \sin \theta}{n}$$

$$d = \frac{\lambda n}{\sin \theta}$$

$$d = \frac{(726 \times 10^{-9}\text{m})(2)}{\sin(0.15^\circ)}$$

$$d = 5.44 \times 10^{-4}\text{m}$$



Path Difference

Constructive interference occurs when n is a whole number (antinodes align). The path difference of two waves which constructively interfere is always a multiple of the wavelength as the waves are not offset.

$$|P_1 - P_2| = n\lambda$$

Destructive interference occurs when n is in the 0.5s (antinodes deconstruct). When there is destructive interference the offset in the two waves is half a wavelength. To find the path difference of 2 waves we need to know at what nodal line the wave is at.

$$|P_1 - P_2| = (n + 0.5)\lambda$$

Two speakers produce 480 Hz, a person walks across parallel to speakers and at the third nodal line where the wave length is 75 cm and the closest speaker is 8m. What is the distance to the other speaker?

$$P_2 - 8m = 3(0.75m) \quad P_2 = 10.25m$$

4.5 (Monday April 17th)

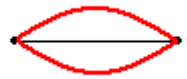
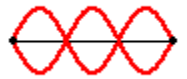
The nature of standing waves

Standing waves are stationary waves as a result of waves being reflected at a resonance frequency creating an amplified stationary wave in which antinodes displace up and down in a stationary location.

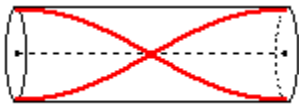
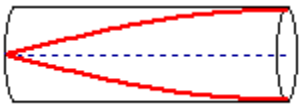
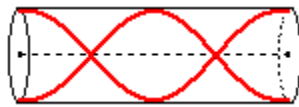
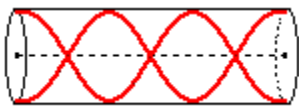
Boundaries and Conditions

Standing waves can also occur in pipes with both ends open, or only one end open. Standing waves can only exist when the waves oscillate at a fundamental frequency at which they resonate. At each harmonic there is a specific frequency that allows for a standing wave.

Standing waves in a string

Harmonic	Frequency	Strings
1st	<i>Fundamental Frequency</i> $= f_1$	 $l = \frac{1}{2}\lambda$
2nd	$f_2 = 2(f_1)$	 $l = \lambda$
3rd	$f_3 = 3(f_1)$	 $l = \frac{3}{2}\lambda$
(n)th	$f_n = n(f_1)$	$l = \frac{n}{2}\lambda$

Standing waves in a pipe

Harmonic	Both Ends Open	One End Open
1st	 $l = \frac{1}{2}\lambda$	 $l = \frac{1}{4}\lambda$
2nd	 $l = \lambda$	None
3rd	 $l = \frac{3}{2}\lambda$	 $l = \frac{3}{4}\lambda$
nth	$l = \frac{n}{2}\lambda$	$l = \frac{n}{4}\lambda$

Note: Antinodes occur only at the open end of the pipe. Pipes with only one open end can only resonate at odd harmonics due to this restriction.

Nodes are points in which the wave does not displace. **Antinodes** are points in which maximum displacement occurs.

Topic 5: Electricity and Magnetism

5.1 Electric Fields

Charge

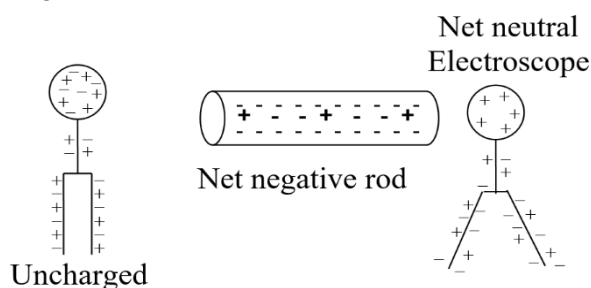
The simple model: All matter is composed of atoms, electric charges are carried by protons and electrons. An **elementary charge** is associated with both these particles, the proton and electron both have a charge magnitude of $1.6 \times 10^{-19} \text{ C}$.

Law of Charges: Like charges repel, opposite charges attract.

Electrical conductivity depends on how tightly the electrons are bound to the nucleus of the atom. A **conductor** is a material that can easily allow for the movement of electrons (delocalized electrons). An **insulator** on the other hand has a strong grasp of its own electrons and can not be easily detached.

Static Theory

Electroscopes: Used to tell if an object is charged. If a charged particle is brought near the head it will repel or attract the electrons within the electroscope causing an **induced charge**. No matter the charge, if a charge is present the leaves will be spread.



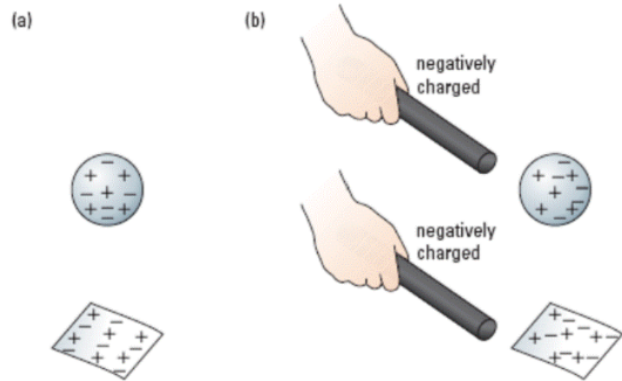
Conservation of Charge: The net charge of an isolated system is conserved

Charging by friction: When materials are rubbed together, electrons can transfer from the material that holds electrons more loosely to the other. **Charge is conserved** as the negative charge lost from the loose object is gained by the tight object.

Charging by conduction: Objects can become charged by the transfer of electrons from simply two objects touching each other. If a charged object touches a neutral object the charge is spread out between the objects according to surface area. When two objects of equal size touch the charge between them will be the same



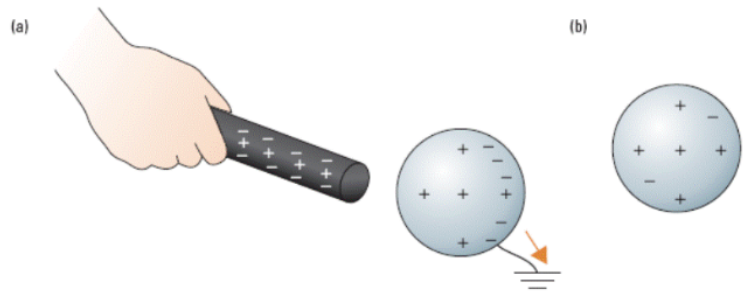
Charging by induction: When a charged object is brought near a neutral object the negative and positive charges within that neutral object are split with one attracted towards the charged object and one charge repelled. This makes each half of the object have a charge.



Ex

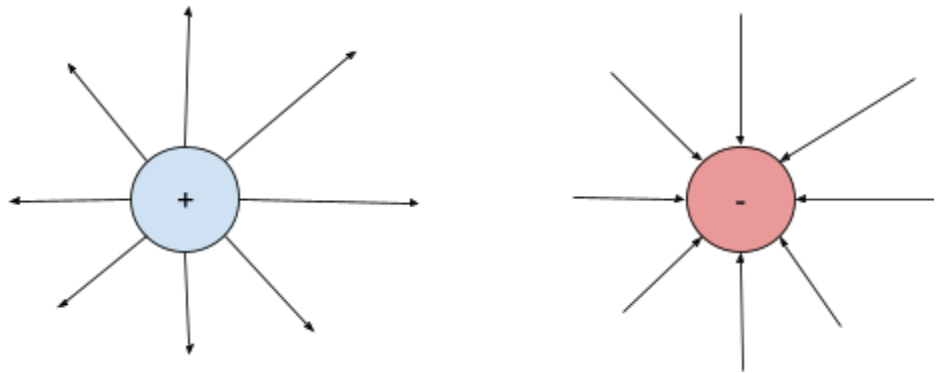
Steps:

1. Negative rod brought near grounded conductor
2. Ground is removed
3. Negative rod is removed
4. Conductor now charged positively.



Electric Field

All charged objects create a field at which other charged objects can interact with. Electric field is a vector and a positive test particle is used to determine the direction of the field. Field lines are evenly spaced and distributed.



In the figure, a positively charged object emits an electric field directed away from the object because a positive test particle would be repelled. And likewise the negative charge has a field pointing towards itself because a positive test particle would be attracted.

$$E = \frac{F_s}{q} \quad E = \text{electric field (N/C)} \quad F_s = \text{electrostatic force (N)} \quad q = \text{charge (C)}$$

Not Given

$$\text{if } F_s = \frac{kq_1q_2}{r^2} \quad \text{and } F_s = Eq \quad \text{then } E = \frac{kq_1}{r^2}$$

Coulomb's Law

The force between two charged objects is determined by the magnitude of charge and separating distance between said two charges.

$$F_s = \frac{kq_1q_2}{r^2}$$

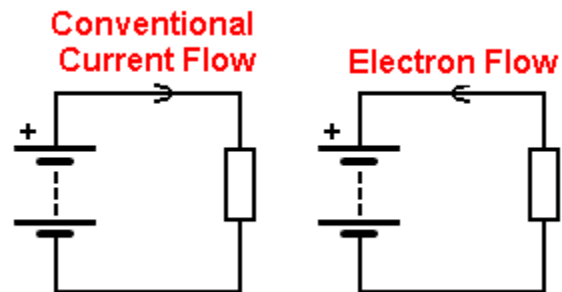
q_1 and q_2 = respective charges k = coulomb's constant r = separating distance

Note: Force is a vector, so when using Coulomb's law only sub in the magnitude of the charges to determine the magnitude of the electrostatic force. Then using the known charges determine if this force is repulsive or attractive, finally add direction according to conventions.

Electric Current

The electrical current through a material is the movement of charge per unit time. Conventional current is the direction of positive flow, and electron flow is the direction of negative flow.

$$I = \frac{\Delta q}{\Delta t}$$



Direct Current and Alternating Current

In circuits the use of direct and alternating current is both required. Direct Current is a current which only flows in one direction. Alternating current is the directional switch of current in a circuit.

Potential Difference/ Voltage

When a charged object is in an apparent electrical field the electric potential difference is the change in electric potential energy per unit charge.

$$V = \frac{W}{q}$$

V = Voltage W = Work done to displace object in field (J) q = charge (C)

Ex. Suppose an electron is accelerated across 120V. Determine the final speed of the electron.

$$120V = \frac{W}{1.6 \times 10^{-19} C} \quad W = 7.5 \times 10^{20} J \quad E_{ps} + E_{ki} = E_{kf}$$

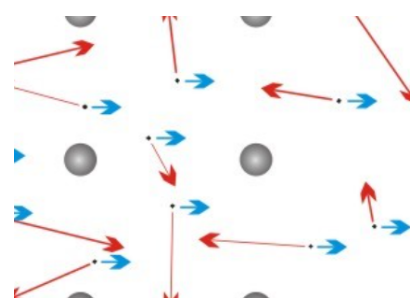
$$7.5 \times 10^{20} J + 0 = \frac{1}{2}mv^2 \quad v = \sqrt{\frac{2(7.5 \times 10^{20} J)}{m}}$$

Drift Velocity

In a current the electrons acted on by a voltage experience of unified velocity propelling them in the direction of current. This speed is called drift velocity.

$$I = nAvq$$

$I = \text{Current (A)}$ $n = \# \text{ of electrons}$ $A = \text{cross sectional area (m}^2\text{)}$
 $v = \text{drift velocity (m/s)}$ $q = \text{charge (C)}$



5.2 (Wednesday April 19th) Heating Effect of Electric Currents

Circuit Diagrams

Resistors: resistors are used to control the current of a circuit with a given voltage. All objects have

Ammeters: Ammeters are used to read the current of a circuit, an ideal ammeter has zero resistance.

Voltmeters: Voltmeters are used to read the voltage over a given portion of a circuit, an ideal voltmeter has infinite resistance and therefore must be wired in parallel.

Emf: Electromotive force is very similar to voltage, but slightly different, the voltage of a battery is the true electric potential of the battery while the emf is the energy/unit charge given at the terminal end of the battery.

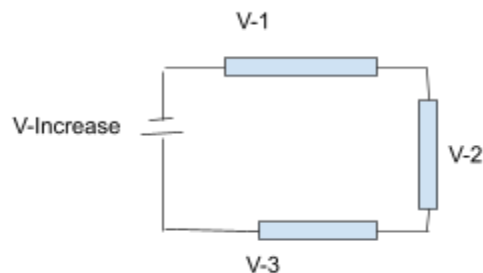
cell		battery	
ac supply		switch	
voltmeter		ammeter	
resistor		variable resistor	
lamp		potentiometer	
light-dependent resistor (LDR)		thermistor	
transformer		heating element	
diode		capacitor	

Kirchhoff's circuit laws

Kirchhoff's Voltage Law

Around any complete path through an electric circuit, the sum of the increases in electric potential (batteries, generators etc) is equal to the sum of decreases in electric potential (resistor, loads).

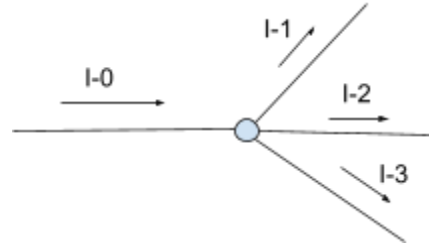
$$V_0 = V_1 + V_2 + V_3 \dots$$



Kirchhoff's Current Law

At any junction point in an electrical circuit, the total electric current into the junction is equal to the sum of current leaving the junction.

$$I_0 = I_1 + I_2 + I_3 \dots$$



Ohm's Law

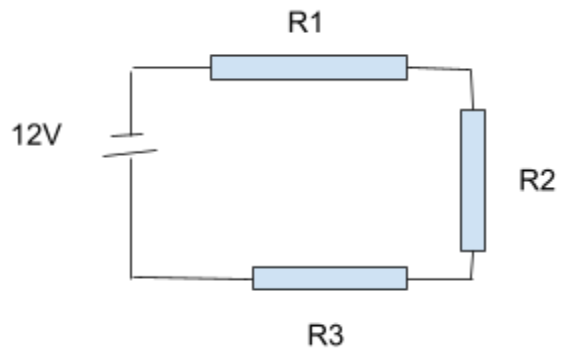
The voltage decrease of a resistor is directly proportional to the current and resistance.

$$V = IR$$

$$V = \text{Voltage (V)} \quad I = \text{Current (A)} \quad R = \text{Resistance (\Omega)}$$

Series Circuit

A series circuit is a circuit loop in which all loads are placed on one path. The effective resistance of a series circuit is the sum of all resistances. In a series circuit the current is constant at all portions.

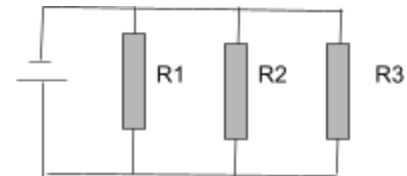


Key Connections:

$$V_{net} = IR_{net} \quad V_1 = IR_1 \quad R_{net} = R_1 + R_2 \dots$$

Parallel Circuit:

A parallel circuit has multiple paths in which loads are wired directly to the power supply individually. Parallel circuits are more efficient but also draw higher current.



Key Connections

$$\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots$$

In a parallel circuit the loads wired in parallel all draw the same amount of voltage from the battery, as each load is on a separate path. So a 12 V battery provides 12 volts to each path of the parallel circuit.

Resistivity

All materials from wires to resistors have a resistivity, as some materials allow the flow of electrons better than others. The resistance provided by a material can be determined by its resistivity constant.

$$R = \rho \frac{l}{A}$$

$$R = \text{Resistance (\Omega)} \quad \rho = \text{resistivity (\Omega m)} \quad l = \text{Length (m)} \quad A = \text{cross sectional area (m}^2\text{)}$$

Power Dissipation

Power dissipated throughout loads is dependent on the resistance, voltage and current. There are multiple ways to find the power dissipated by a load as voltage and current and resistance are all connected.

$$P = IV \quad P = I^2 R \quad P = \frac{V^2}{R}$$

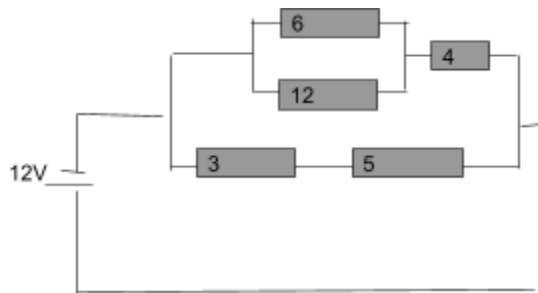
$P = \text{Power (W)}$ $I = \text{Current (A)}$ $V = \text{Voltage (V)}$ $R = \text{Resistance } (\Omega)$

Complex Circuits

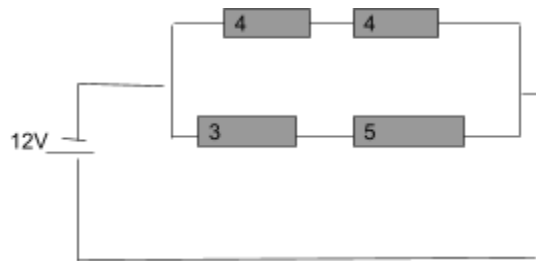
Complex circuits contain loads in both series and parallel. The best way to tackle complex circuits is to re-write circuits and simplify using effective resistances.

Ex

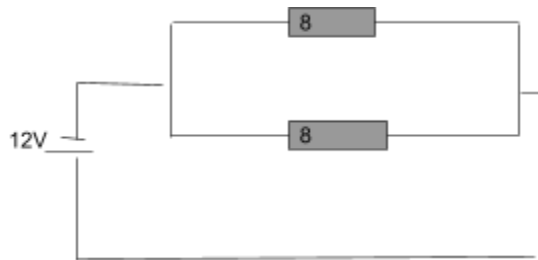
Initial Circuit



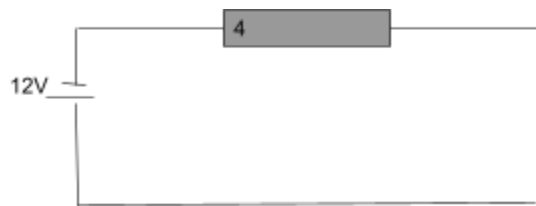
First Redraw



Third Redraw



Final Redraw



After simplifying by finding the effective resistances of parallel portions and coming up with the final redraw, work your way up using Ohm's Law and Kirchoff's current and voltage laws to find required magnitudes.

Kirchhoff's Loop Rule

In circuits with multiple loops and sources, if possible, find two loops that provide current in the same direction. Using Kirchhoff's current law label and set all currents according to junctions and direction. Finally create a system of equations using ohm's law for both loops.

Ex Find the current in the $4\ \Omega$ resistor. Specify direction. Then find the power supplied by the $9.0\ \text{V}$ power supply.

Kirchhoff's Current Law: $I_1 + I_2 = I_3$

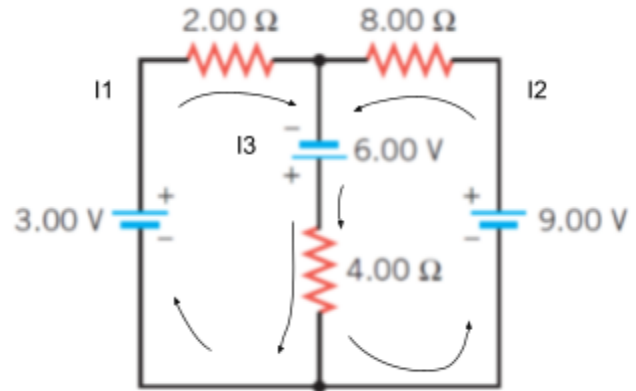
Kirchhoff's Voltage Law: $V_{up} = V_{down}$

Loop 1: $9 + 6 = 8I_2 + 4I_3$

Loop 2: $3 + 6 = 2I_1 + 4I_3$

Loop 1: $9 + 6 = 8I_2 + 4I_3$

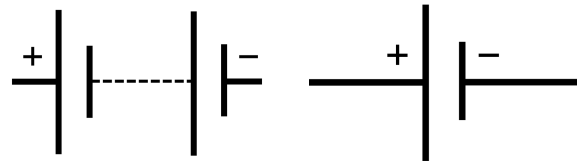
Loop 2: $3 + 6 = 2(I_3 - I_2) + 4I_3$



5.3 Electric Cells

Cells

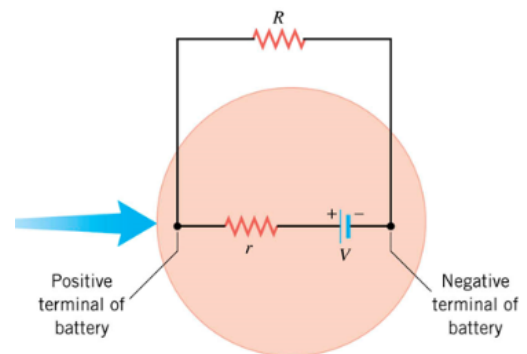
A cell in a circuit acts as a source of electrical energy and creates an electric potential difference at its terminals. A battery is made up of two cells.



Internal Resistance

The resistance within the battery or cell is determined by the material it is made out of. The **terminal voltage** is the voltage supplied at the terminals of the battery. The **electromotive force** is the original potential difference provided.

$$\varepsilon = I(R + r) \quad V_t = \varepsilon - Ir$$



5.4 Magnetic Effects of Electric Currents

Domain Theory of Magnetism

All ferromagnetic substances are composed of a large number of regions called “magnetic domains”. Each domain behaves like a tiny bar magnet. Normally when a magnetic field isn’t present they are scattered. But whenever in the presence of a magnetic field the domains align and the substance becomes a temporary magnet.

Law of Magnetism

Magnets have two poles in which like poles repel and opposite poles attract.

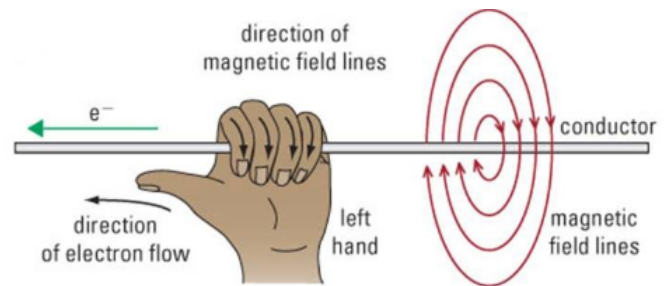
Magnetic Field

If an object has distinct North and South poles the object has a magnetic field. Field lines never cross and always go from the north pole to south pole. Magnetic fields are caused by the presence of magnets or moving charges. The unit is T, Tesla.

Electromagnetism and Hand Rules

Magnetic Field from Wire Carrying Current

It was observed that when a wire had a flow of current through it a magnetic field $\vec{B}$ is created.



1st Hand Rule

Point your thumb in the direction of flow (current/e flow) and curl your fingers. The fingers show the direction of the magnetic field around a wire.

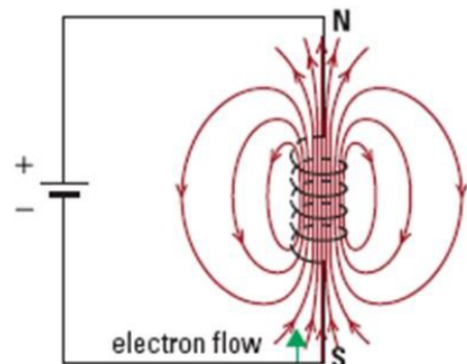
Note: Use your left thumb when pointing in the direction of electron flow and right thumb for conventional current.

Magnetic Fields Present in a Solenoid

When the wire with current is repeatedly coiled around a hollow tube a solenoid is created. As each wire contains its surrounding magnetic field the fields add up to create a magnet with 2 poles.

2nd Hand Rule

Curl your fingers in the direction of flow (left for electron flow, right for conventional current) around the solenoid. Your thumb will always point to the North Pole created by the solenoid..



Current Carrying Wire and External Magnetic Fields

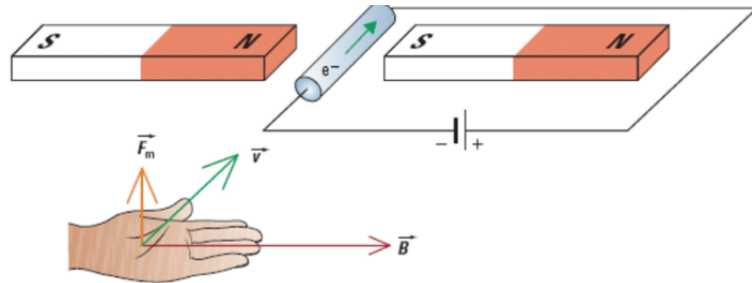
Current Carrying wires and the magnetic fields produced can interfere with external magnetic fields present.

3rd Hand Rule: Point thumb in direction of flow (left for electron flow, right for conventional current), direct your fingers in the direction of the external magnetic field. The direction at which your palm faces is the direction of magnetic force.

Formula for Magnetic Force

$$\vec{F}_m = I\vec{l}\vec{B}$$

Direction must always be considered.



Charged Particle in Magnetic Fields

It was determined that a wire carrying a current interacts with external magnetic fields to produce a force. This force comes from the cumulative effects of the forces on the individual charges within the wire.

Consider: $\vec{F}_m = I\vec{l}\vec{B}$ $I = \frac{q}{t}$ $l = vt$ $\rightarrow$ $\vec{F}_m = \frac{q}{t}(vt)\vec{B}$

By substituting current and length with variables related to individual charges a formula can be made that relates the magnetic force acting on a particle in motion.

$$\vec{F}_m = qv\vec{B}$$

$$\vec{F}_m = \text{magnetic force (N)} \quad q = \text{charge (C)} \quad v = \text{speed (m/s)} \quad \vec{B} = \text{magnetic field (T)}$$

Note: The third hand rule applies to particles, if the charge is negative use left hand and if charge is positive use right hand. Thumb points in the direction of motion.

Magnetism and Uniform Circular Motion

When a particle of speed is shot into a uniform magnetic field the magnetic force acting will change the motion of the particle putting it into temporary circular motion.

Combining with circular motion formulae:

$$qv\vec{B} = \frac{mv^2}{r}$$



Topic 6: Circular Motion and Gravitation

6.1

Centripetal Force and Acceleration

The net force keeps an object in uniform circular motion. The force is always directed toward the center of motion.

Formulae

$$T = \frac{1}{f} \quad \left| \vec{a}_c \right| = \frac{v^2}{r} \quad \left| \vec{a}_c \right| = \frac{4\pi^2 r}{T^2} \quad \left| \vec{F}_c \right| = \frac{mv^2}{r} \quad \left| \vec{F}_c \right| = \frac{m4\pi^2 r}{T^2} \quad \vec{F} = ma$$

Horizontal Circular Motion

Centripetal force is a singular force acting on an object. Ex constant frictional force keeping a car in a corner.

A 1250 kg car enters an unbanked curve of 80 m radius. If the coefficient of friction between the pavement and the car tires is $\mu_s = 0.80$, and $\mu_k = 0.72$ what is the maximum speed which the car can go around the curve?

$$F_c = F_f$$

$$F_f = \mu F_n$$

$$\mu F_n = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{\mu F_n r}{m}}$$

$$v = \sqrt{\frac{(0.8)(9.81)(1250)(80)}{(9.81)}}$$

Vertical Circular Motion

Force centripetal is the net force of gravity and another acting on an object. Ex Roller Coaster (Force Normal and Gravity).

In vertical circular motion there are two scenarios, when the two forces are in the same direction and when the two forces are opposite. Force centripetal could be the difference of two or the sum of two, but the magnitude of Centripetal Force is always constant.

Ex. a 300g mass is on a 45 cm string swung in a vertical circle with a period 1.2 seconds.
Determine the tension force at the top and bottom

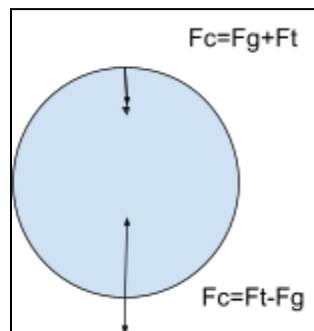
At Top: (Force Gravity and Force Tension in same direction)

$$\vec{F}_c = \vec{F}_g + \vec{F}_T$$

$$\vec{F}_g + \vec{F}_T = \frac{m4\pi^2 r}{T^2}$$

$$\vec{F}_T = \frac{m4\pi^2 r}{T^2} - \vec{F}_g$$

$$\vec{F}_T = \frac{(0.3kg)(4)\pi^2(0.45m)}{(1.2s)^2} - (0.3kg)(9.81)$$



At Bottom: (Force Gravity and Force Tension in opposite directions)

$$\vec{F}_c = \vec{F}_T - \vec{F}_g$$

$$\vec{F}_T = \frac{(0.3kg)(4)\pi^2(0.45m)}{(1.2s)^2} + (0.3kg)(9.81)$$

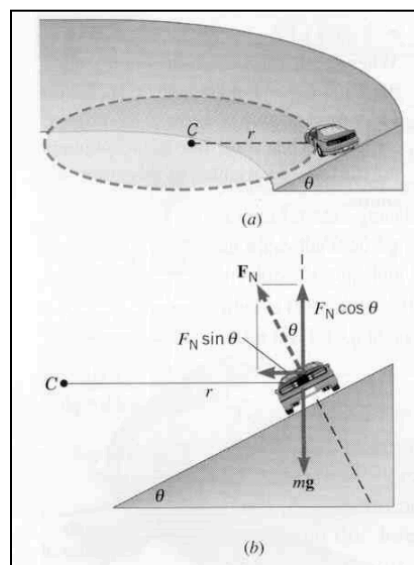
Banked Corners

When an object turns a corner that is angled the horizontal component of the normal force acts as the centripetal force.

$$F_c = (mg) \tan \theta$$

$$(mg) \tan \theta = \frac{m4\pi^2 r}{T^2}$$

$$(mg) \tan \theta = \frac{mv^2}{r}$$



Rotational Kinematics

Arc Length

$$\theta = \frac{s}{r} \quad \theta = \text{angle in radians} \quad s = \text{arc length} \quad r = \text{radius}$$

Angular Velocity

$$\omega = \frac{\Delta\theta}{t} \quad \omega = \frac{2\pi}{T} \quad v_t = r\omega$$

Ex I drive at 110 km/h in my car, my car tires are 0.74 m in diameter. What is the angular velocity of the wheels in radians/sec and revolutions/min.

$$v_t = r\omega$$

$$\omega = \frac{v_t}{r}$$

$$\omega = \frac{110 \text{ km/hr} \times 1000 \text{ m} / 3600 \text{ s}}{0.37 \text{ m}}$$

$$\omega = 82.6 \text{ rad/s}$$

$$\frac{82.6 \text{ rad}}{\text{s}} \times \frac{60 \text{ s}}{2\pi} = 788.7 \text{ rpm}$$

Combining Angular Velocity

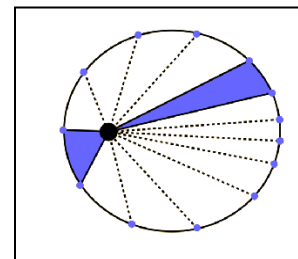
$$v_t = r\omega \quad + \quad \left| \vec{F}_c \right| = \frac{mv^2}{r} \quad \rightarrow \quad \left| \vec{F}_c \right| = m\omega^2 r$$

6.2

Gravitation and Gravitational Field Strength

Kepler's Law

1. The orbit of a planet is an ellipse with the sun at one of the foci.
2. A line that connects a planet to the sun sweeps out equal areas in equal times
3. The square of the orbital period of any planet is directly proportional to the cube of the semimajor axis (radius) of its orbit.



$$T^2 \propto r^3$$

$$\text{If } T^2 \propto r^3 \text{ for any given planet then: } \frac{r_1^3}{T_1^2} = \frac{r_2^3}{T_2^2}$$

Where the ratio of one planet's orbital radius and period is equal to another planet's orbital radius and period.

Universal Law of Gravitation

All objects in the universe attract each other with equal and opposite forces.

$$F_g = G \frac{m_1 m_2}{r^2}$$

F_g = Force of gravity

G = Universal Gravitational Constant = $6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

r = radius

$m_{1 \text{ and } 2}$ = mass of 2 objects

The force of gravity is the force of attraction 2 given objects have, if one object attracts another with a force, then the other attracts it as well with equal and opposite force.

Gravitational Field Strength

$$g = \frac{Gm}{r^2} \quad F_g = mg$$

Every object has a gravitational field. The force another object feels is inversely proportional to the square of the distance between them.

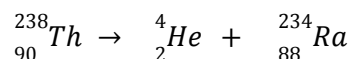
Topic 7: Atomic, Nuclear and Particle Physics

7.1

Radioactive Decay

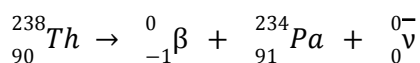
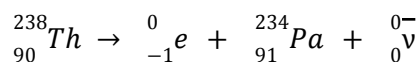
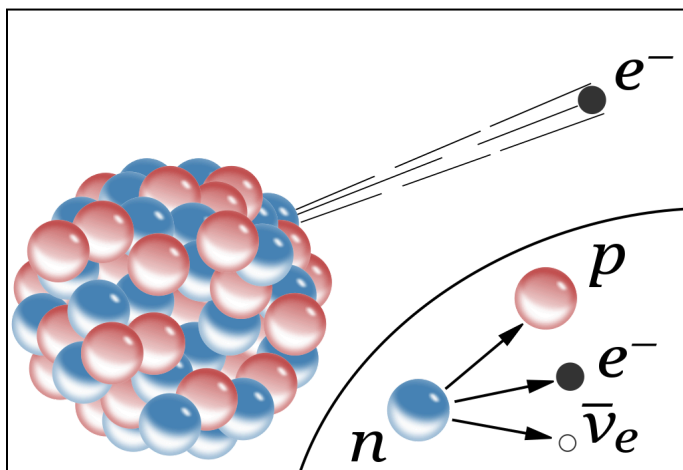
Alpha Decay (α^{2+}) : The ejection of a high speed helium nucleus.

- They are the nucleus of helium atoms ${}^4_2\text{He}$.
- They are ejected at a high speed but have a range of only a centimeter in air.
- An ordinary sheet of thin aluminum foil or paper stops them.
- Can be deflected by magnetic and electric fields.
- Very dangerous if ingested/inhaled.



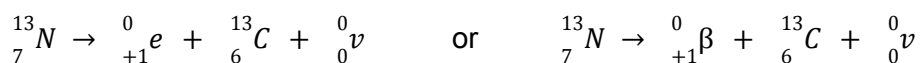
Beta Negative Decay (β^-): Emission of high energy electrons and antineutrino.

- Beta particles are formed when a neutron changes into a proton emitting a high energy electron and an antineutrino.
- Able to penetrate several millimeters of aluminum
- Can be deflected by magnetic and electric fields.



Beta Positive Decay (β^+): Emission of a high-energy positron and a neutrino

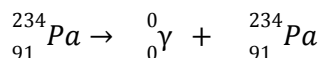
- Beta positive particles or positrons are formed when a proton changes into a neutron in the nucleus emitting a high energy anti-electron and neutrino
- Same properties as a beta negative particle except oppositely charged.



Weak Nuclear Force: In Beta the transformation of a neutron into a proton involves a fundamental force called the weak nuclear force. It acts on the electrons and neutrinos catalyzing the change.

Gamma Rays (γ): The emission of a high-energy photon

- Wavelengths and energies can vary.
- Charges within the nucleus redistribute themselves
- High Energy gamma rays can penetrate at least 30 cm of lead or 2 km of air
- Cannot be deflected by electric and magnetic fields.



Pair Production: When matter is created an antimatter particle is also created with identical mass and opposite charge.

Absorption Characteristics

Radiation	Typical Penetration	Ionization	Hazard
Alpha	5 cm air. No skin Pen	high	low
Beta	30-50 cm air. 1 cm skin	moderate	low
Gamma	Penetrates through body	low	High

Half Life

The time for half of the atoms of a given quantity of radioactive isotope to decay is the half-life of that isotope.

Definitions:

1. The decay rate is known as the **activity** of the radioactive substance. The units for activity is the becquerel (Bq) 1Bq = 1 decay/second.
2. **Background activity**: The natural radiation in the area that is not directly related to the radioactive sample.

$$N = N_0 \left(\frac{1}{2} \right)^n \quad n = \frac{t}{t_{0.5}}$$

N = Amount of substance left

N_0 = Amount of substance

n = halflife exponent

t = time

$t_{0.5}$ = halflife duration

7.2

Mass Defect and Binding Energy

When scientists compared the calculated or theoretical mass for an isotope with measured mass, the expected values were not identical. This difference in mass is the **mass defect** Δm .

In general $\Delta m = m_{\text{Higher Energy}} - m_{\text{Lower Energy}}$

$$\Delta m = m_{\text{Nucleons}} - m_{\text{Nucleus}}$$

$$\Delta m = m_{\text{Radioactive}} - m_{\text{Decay Product}}$$

Because there is a **strong nuclear force** holding the nucleus together it is more stable than its parts (proton and neutron). Therefore the energy required to break the strong nuclear force is absorbed on the nucleons and marginally greater than the nucleus.

$$E = \Delta mc^2$$

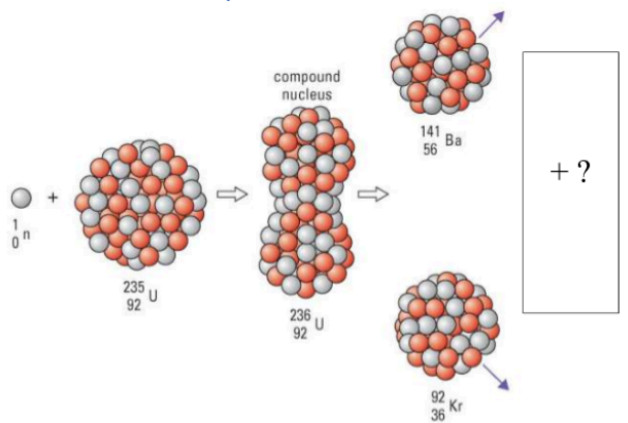
Unified Atomic Mass Unit: 1 u = 1/12 mass of carbon

Fission and Fusion

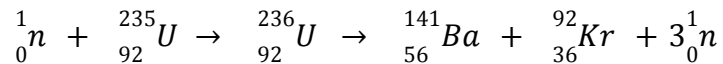
Fission: When a nucleus with $A > 120$ splits into smaller nuclei, each product has a greater binding energy per nucleon than the original nucleus.

Problems: Highly radioactive by-products. Fear of nuclear meltdown. Huge amount of electrical energy per kg of fuel used.

Example Problem:



Chemical Equation:



$$92p + 36p + 56p \quad 143n = 85n + 56n + 3n$$

The excess neutrons are used to chain fission reactions.

$$\text{U-235} = 235.043930u$$

$$\text{Neutron} = 1.008665u$$

$$\text{Ba-141} = 140.914412u$$

$$\text{Kr-92} = 91.926159u$$

How much energy is produced?

$$\Delta m = m_{in} - m_{out}$$

$$\Delta m = 0.186034u$$

$$E = \Delta mc^2$$

$$E = 0.186034u (931.5 \text{ MeV } u^{-1} c^{-2})(c^2 u)$$

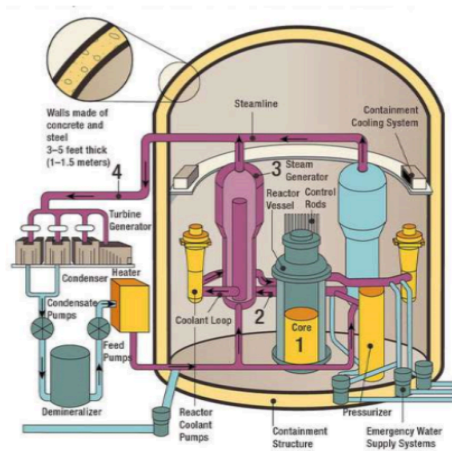
$$E = 173.6 \text{ MeV}$$

Power Reactors:

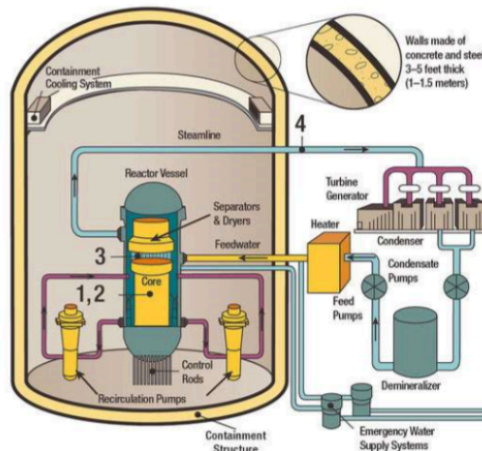
Pressurized Water: In a PWR the primary coolant (superheated water) is pumped under high pressure to the reactor core, then the heated water transfers thermal energy to a steam generator. In contrast to a boiling water reactor, pressure in the primary coolant loop prevents the water from boiling within the reactor.

Boiling Water: The other common US power reactor type is the Boiling Water Reactor. As the name implies, the inner loop contains both liquid water and steam, which drives the turbines. Since this inner water will contain radioactive material, this design can cause contamination of turbine blades.

Pressurized Water Reactor:



Boiling Water Reactor:



Fusion: When two low-mass nuclei combine to form a single nucleus with $A < 60$, the resulting nucleus is more tightly bound. This fusion reaction gives off energy equal to the difference between the total binding energy of the original nuclei and the binding energy of the product.

Benefits: More energy released per mass of fuel used. There is an abundance of easily obtained fuel. Zero CO₂ emissions. No radioactive by-products.

7.3 Structure of Mater

Quarks

Hadrons

Hadrons exist within the nucleus and therefore interact with strong nuclear forces.

There are three types of hadrons: Baryons, Antibaryons and Mesons. Baryons are composed of 3 quarks qqq , antibaryons are composed of 3 antiquarks $\overline{qqq}$. Mesons are made of one quark and an antiquark. $q\overline{q}$.

Generation	Name	Symbol	Baryon Number	Charge
First	Up	u	$1/3$	$+\frac{2}{3}e$
	Down	d	$1/3$	$-\frac{1}{3}e$
Second	Charm	c	$1/3$	$+\frac{2}{3}e$
	Strange	s	$1/3$	$-\frac{1}{3}e$
Third	Top	t	$1/3$	$+\frac{2}{3}e$
	Bottom	b	$1/3$	$-\frac{1}{3}e$

Protons and Neutrons are Baryons made out of three first generation quarks. The anti matter of a proton and neutron then must be antibaryons and can only be made of 3 antiquarks.

$$Proton = +1e = uud = \frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e$$

$$Neutron = 0 = udd = \frac{2}{3}e - \frac{1}{3}e - \frac{1}{3}e$$

When a neutron changes to a proton in β^- Decay the udd neutron changes to a uud proton.

Proving conservation of charge using quark notation.

Beta Negative Decay:

$${}^A_ZX \rightarrow {}^A_{Z+1}Y + {}^0_{-1}e + {}^0_0\overline{\nu}$$

$$udd \rightarrow uud + {}^0_{-1}e + {}^0_0\overline{\nu}$$

Conservation Baryon Number

Each quark has a Baryon number of $\frac{1}{3}$ and each anti-quark has a baryon number of $-\frac{1}{3}$. For reactions to happen the baryon number must be conserved.

Conservation of Strangeness

All quarks have a strangeness number of zero except the strange quark, s . The strange quark has a strangeness number of -1, and the anti-quark $\bar{s}$ has a strangeness number of 1. All that involves the strange quark must have balanced strangeness.

Quark Combinations: $K^- = \bar{s}u$ $K^0 = d\bar{s}$ $K^+ = u\bar{s}$ $X = \text{Unknown}$

$$K^- + p \rightarrow K^0 + K^+ + X$$

Quark Notation: $\bar{s}u + uud \rightarrow d\bar{s} + u\bar{s} + X$

Charge: $(-\frac{1}{3} - \frac{2}{3}) + (\frac{2}{3} + \frac{2}{3} - \frac{1}{3}) \rightarrow (-\frac{1}{3} + \frac{1}{3}) + (\frac{2}{3} + \frac{1}{3}) + X$

Baryon Number: $0 + 1 \rightarrow 0 + 0 + X$

Strangeness: $-1 + 0 \rightarrow 1 + 1 + X$

X must have charge of -1 , Baryon Number of 1, and strangeness of -3

A baryon number of 1 means particle X is made of 3 quarks and is either a baryon or anti-baryon.

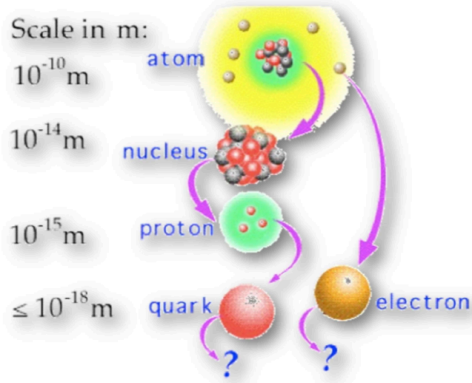
$X = sss$, Strangeness: -3 , Baryon #: 1, Charge: -1

sss satisfies all conservation laws therefore is correct.

Mediating Particle: A virtual particle that carries one of the fundamental forces, mediating particles with limited range are **bosons**. ($1\text{fm} = 10^{-15}\text{m}$)

Force	Range	Relative Strength to proton in nucleus	Mediating Particle	Particle Observed?
Electromagnetic	Infinite	10^{-2}	Photon	Yes
Weak Nuclear	$<0.003\text{ fm}$	10^{-6}	W^-, W^+, W^0	Yes
Strong Nuclear	$<1\text{ fm}$	1	Gluon	Indirectly
Gravity	Infinite	10^{-38}	Graviton	No

Particle Scale Diagram



Quark Confinement:

A baryon or meson can never be split into its individual quarks.

Because of the strong nuclear force binding quarks together, quarks behave like they have an elastic band attached to each other, and no matter how much energy is added the quarks will always reattach.

Leptons

All leptons have a lepton number of +1 and all anti-leptons have a lepton number of -1.

Generation	Name	Symbol	Charge	Lepton Number
First	Electron	e	- 1	1
	Electron Neutrino	ν_e	0	1
Second	Muon Particle	μ	- 1	1
	Muon Neutrino	ν_μ	0	1
Third	Tau Particle	τ	- 1	1
	Tau Neutrino	ν_τ	0	1

-Leptons unlike Quarks do not react to gluons

-Quarks react to both gluons and photons

-Leptons and quarks also react to gravitons

Leptons are always conserved by both **generation** and **charge**

Check if each equation is feasible:

	$n + p \rightarrow \mu^+ + \nu_\mu$	$p \rightarrow n + e^+ + \nu_e$
Baryon #:	$1 + 1 \rightarrow 0 + 0$	$1 \rightarrow 1 + 0 + 0$
Lepton #:	$0 + 0 \rightarrow (-1_2) + 1_2$	$0 \rightarrow 0 + (-1_1) + (1_1)$
Feasible:	No (Baryon # not conserved)	Yes

Antimatter

Quantum theory predicts that each kind of ordinary particle has a corresponding antiparticle. One of the startling properties of antimatter is that a collision between a particle and its antiparticle can **annihilate** both particles and create a pair of high energy gamma-ray photons.

Electron-Positron Annihilation

$$e^{-} + e^{+} \rightarrow 2 \gamma$$

$$E = (2 \times 0.511 \text{ MeV}) c^2$$

$$\lambda = \frac{hc}{E}$$

$$\lambda = \frac{1.24 \times 10^{-6} \text{ eV m}}{(1.022 \text{ MeV} \times \frac{10^6 \text{ eV}}{1 \text{ MeV}}) 0.5}$$

$$\lambda = 2.43 \times 10^{-12} \text{ m}$$

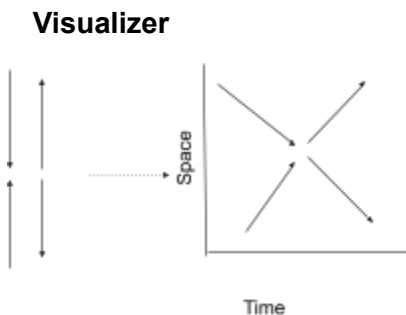
Note: Remember to balance, the energy created by the collision is the energy of 2 photons, to find the wavelength of the photon, energy must be divided by 2.

Gamma Ray photons travel at the speed of light, because photons are massless, momentum is 0.

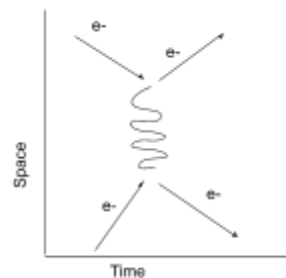
Feynman Diagrams

Two Colliding Electrons

Consider: Two electrons colliding and repelling just with vectors, when you add time and space axis it looks as such.



Correct Feynman



Notation:

- Particles are represented with straight arrows.
- Exchange (force) particles are shown with wiggly lines (mediating particles)
- Even though the arrows don't both point together, the arrow pointing forward in time is a regular particle and the arrow pointing in the opposite direction of time is an antiparticle.

Particle →

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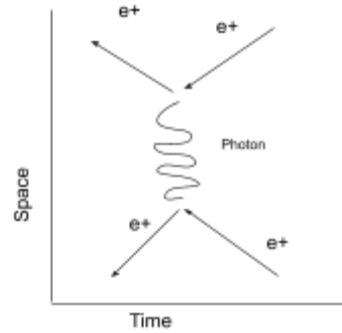
Electromagnetic and weak exchange

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Strong exchange

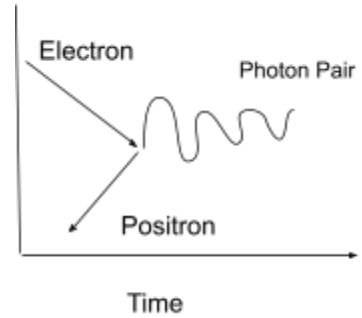
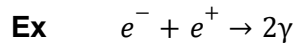
Positron Repulsion

Positrons and all antimatter have arrows directing backwards in time, the particle isn't actually traveling backwards in time the arrow direction only notes the type of particle, direction is always left to right.



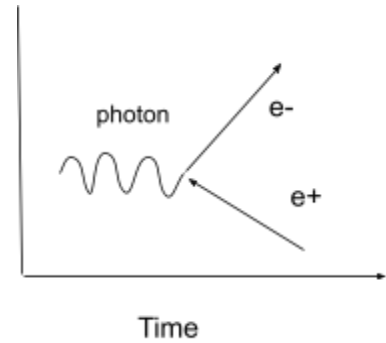
Annihilation

When a particle and its counter antiparticle collide to create high energy gamma-ray photons.

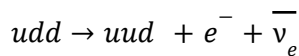


Pair Production

The production of a particle and its antiparticle from a neutral boson (particle carrying force). Just like how matter and antimatter can annihilate into a neutral boson, a high energy boson can also create a pair of matter.

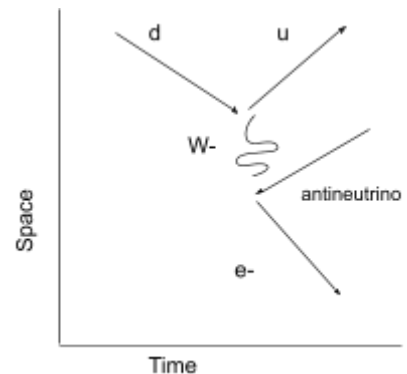
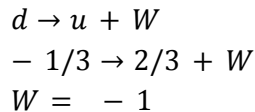


Flavor Change



Down quark undergoes flavor change into up quark the charge conserved through this interaction is present in the boson W^-

Why is the W negative?

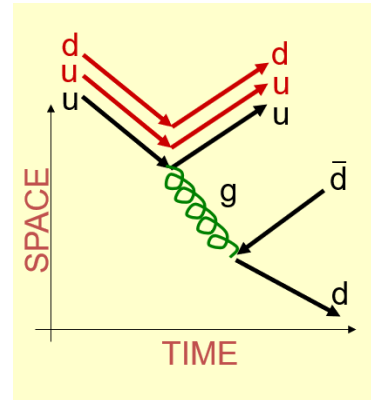


Explain what is happening in the Feynman Diagram.

An up quark in a proton (uud) emits a high energy gluon which then pair produces a down and anti-down quark

$$u \rightarrow g + u \quad g \rightarrow d + \bar{d}$$

Charge: $2/3 \rightarrow 0 + 2/3 \quad 0 = -1/3 + 1/3$



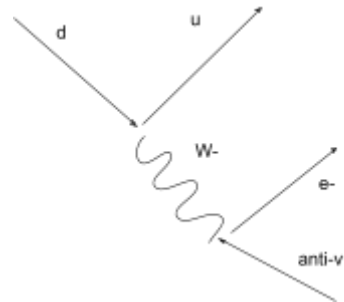
Fully explain and write/draw out the process of a beta negative emission.

Nuclear Notation $n \rightarrow p + e^- + \bar{\nu}_e$

Quark Notation: $udd \rightarrow uud + e^- + \bar{\nu}_e$

Charge: $(2/3) + (-1/3) + (-1/3) \rightarrow (1) + (-1) + (0)$

Weak Interactions: $d \rightarrow u + W^- \quad W^- \rightarrow e^- + \bar{\nu}_e$



Topic 8: Energy Production

8.1 Energy Sources

Specific Energy and energy density of fuel source

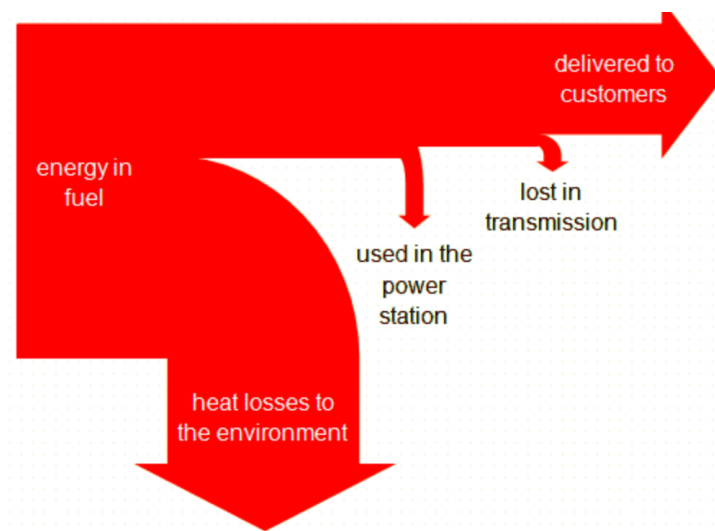
Specific Energy: The specific energy of a substance is the energy available per unit mass
Energy/Mass. (J/kg)

Energy Density: The energy density of a substance is the energy available per unit volume
Energy/Volume.(J/m³)

Sankey Diagrams

According to the second law of thermodynamics when energy is converted to work some of the energy is lost due to the process.

Sankey Diagrams show the loss of energy and the useful energy out. Coming from the left is the input energy through a process, all energy lost due to some method is shown by directing energy down, the useful energy then continues to the next process.



Primary Energy Sources

A primary energy source is a source of energy that has not been subjected to any conversions or transformation processes.

Secondary Energy Sources

Secondary energy sources are sources which have been converted from one energy type to another, the most popular example is electricity, which virtually powers all appliances but is not a primary source.

Renewable Energy

Energy that can be replaced within a reasonable amount of time (or can not be depleted)

Sun-Derived Energy Sources	
Fuels • Coal • Oil • Gas • Wood • Biomass	Non-fuels • Solar Energy • Hydroelectric • Wind • Wave • Photovoltaic
Non-Sun-Derived Energy Sources	
Fuels • Hydrogen • Uranium • Chemical	Non-fuels • Tidal • Geothermal • Chemical

Theory

- Renewable energy is better than non-renewable
- Oil and Gas are better than coal because they burn more efficiently and have lower carbon dioxide emissions.
- Coal is cheaper than oil and gas
- Nuclear has no carbon dioxide emissions
- Hydroelectric systems can store energy by using the generated energy to pump the water back up hence recycling the loop.
- Nuclear waste lasts for thousands of years
- Wind turbines and photovoltaic cells depend on weather and have low output

Wind Turbines

Derivation

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}\rho A v t v^2 = \frac{1}{2}\rho A t v^3 \quad A = \pi r^2$$

The mass of air is the density ρ multiplied by volume Avt where A is the cross sectional area and vt is the distance air has traveled in a second.

$$P = \frac{E}{t} = \frac{1}{2}\rho A v^3$$

Solar Heating Panels vs Photovoltaic Cells

Solar heating panels are dark panels that absorb sunlight and convert it directly into heat. There is no storage of energy, just the transfer.

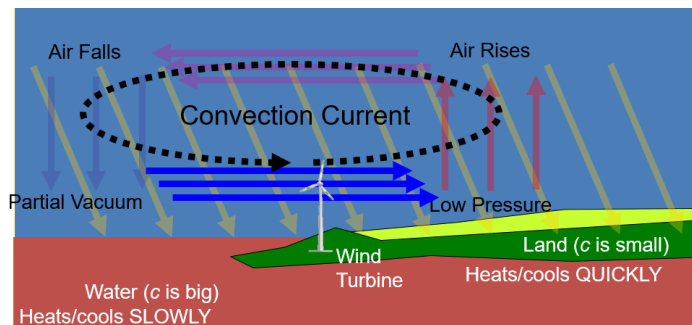
Photovoltaic Cells convert sunlight directly into electricity. These cells can only be so efficient therefore it is very hard to efficiently set up solar energy.

8.2 Thermal Energy Transfer

Conduction and Convection

Conduction: The transfer of thermal energy through direct contact. The vibrations of the higher heated object transfer through a conductive medium or by contact until both objects are at thermal equilibrium

Convection: The flow of thermal energy from mass and molecular movement. As more heat is present in gasses the air density becomes less, this causes the air to rise, as the air cools it then falls back down. This loop is called a convection current.



Black Body Radiation

Solids can absorb the most electromagnetic radiation out of all the states of matter. Depending on the color of the solid it can be determined which wavelengths are reflected or absorbed. The color of an object is what wavelength is reflected (grass is green, it reflects green wavelengths). White reflects all wavelengths and black absorbs all wavelengths.

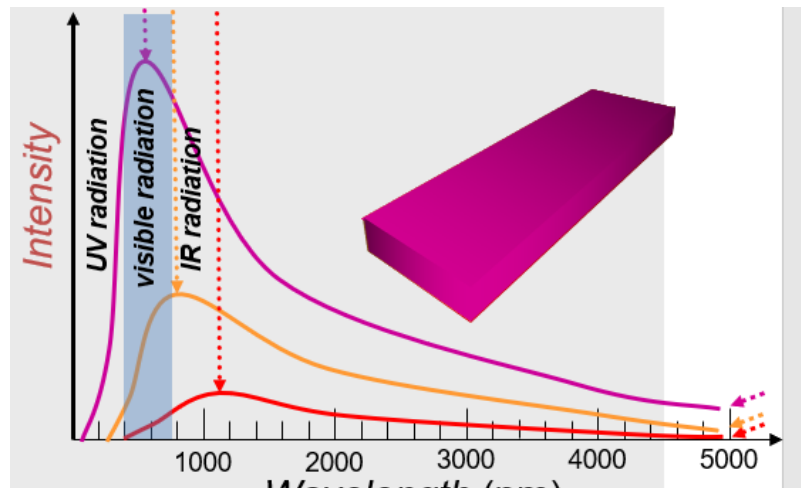
Black absorbs all wavelengths it comes into contact with, this includes both visible and other light. These wavelengths are then trapped, and in a perfect black body all wavelengths are emitted.

As we heat solids to incandescent states it can be observed at different temperatures the intensity of the types of wavelengths differ.

Two Major Trends:

The higher the temperature the greater intensity of all wavelengths.

The higher the temperature the shorter the wavelength of most emitted light.



Wien's Displacement Law

Wien's Displacement law shows the most intense wavelength of light at a given temperature.

$$\lambda_{max} = \frac{2.90 \times 10^{-3}}{T}$$

λ_{max} = wavelength of greatest intensity (m) T = Temperature (K)

Stefan Boltzman Law

The Stefan-Boltzman Law relates the power emitted by a black body's surface area to its temperature.

$$P = \sigma A T^4$$

P = Power (W) σ = Stefan Boltzman Constant A = Surface Area (m^2) T = Temperature (K)

Note: Use the average temperature of an object when using the stefan boltzmann law.

EXAMPLE: Mercury has a radius of 2.50×10^6 m. Its sunny side has a temperature of 400°C (673 K) and its shady side -200°C (73 K). Treating it like a black-body, find its power.

SOLUTION: $P = \sigma AT^4$

$$P = \sigma(4\pi(2.50 \times 10^6 \text{ m})^2)\left(\frac{673+73}{2}\right)^4 \quad P = 8.6 \times 10^{16} \text{ W}$$

Solar Constant: $1.36 \times 10^3 \text{ W/m}^2$

The solar constant is the average amount of power earth receives from the sun per meter square.

Emissivity

The emissivity of an object e is a number between 0 and 1 that quantifies the emission and absorption properties of a body. A perfect black body has an emissivity of 1 as it absorbs and emits all energy.

$$e = \frac{P_{\text{body}}}{P_{\text{black body}}}$$

Albedo

A body's albedo quantifies its ability to **scatter** energy, an object like a mirror which reflects almost all energy has an albedo of 1. Albedo is the ratio of power reflected over power from source. If 30 percent of the source power is reflected 70 percent must be received/absorbed.

$$\text{Albedo} = \frac{P_{\text{scattered}}}{P_{\text{incident}}}$$

EXAMPLE: Assuming an albedo of 0.30, find, for Earth: The radius of Earth is $r = 6.37 \times 10^6$ m

(a) the power of the sunlight received.

$$\text{Power Received} = \pi(6.37 \times 10^6 \text{ m})^2 \times \frac{1.36 \times 10^3 \text{ W}}{\text{m}^2} \times 0.7 = 1.21 \times 10^{17} \text{ W}$$

b) the *predicted* temperature due to the sunlight reaching it.

Power Absorbed = Power Emitted Thermal Equilibrium-No temp change

$$P = \sigma AT^4 \quad 1.21 \times 10^{17} \text{ W} = \sigma 4\pi r^2 T^4$$

$$T = \sqrt[4]{\frac{1.21 \times 10^{17} \text{ W}}{\sigma 4\pi (6.37 \times 10^6 \text{ m})^2}}$$

$$T = 254 \text{ K or } -19^\circ\text{C}$$

The greenhouse effect

Greenhouse gasses readily absorb infrared light from the photons of energy striking the earth due to natural resonance frequencies of molecules. Molecules can hold on infrared radiation by vibrating, translating and rotating.

Option C: Optics

C.1 Introduction to Imaging

Real and Virtual Images

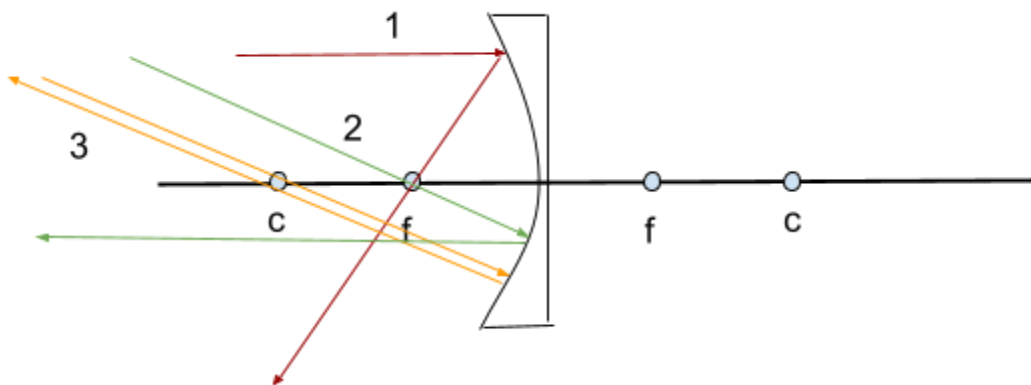
Images formed by mirror and lenses can be categorized into **real** and **virtual**.

Real Images can be projected and “touched”, formed by the intersection of real light rays. The real image of an erect object is always inverted.

Virtual Images exist behind or “within” the mirror. Can not be “touched” and is the result of the intersection of virtual light rays.

Mirrors

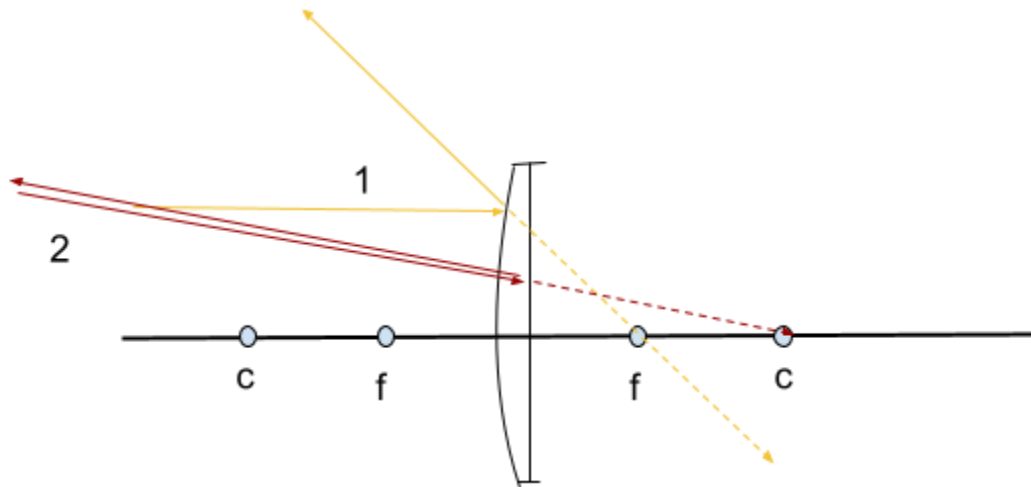
Concave Mirror: A concave mirror has converging properties and will reflect light rays to the normal line.



Key Rays:

1. Rays parallel to the principal axis will always converge into the focal point of a concave mirror.
2. Rays that are incident through the focal point will always reflect parallel to the principal axis.
3. Rays that are incident through the center of curvature will always be reflected directly back.

Convex Mirror: A convex mirror has diverging properties and will reflect light waves away from the normal.



Key Rays

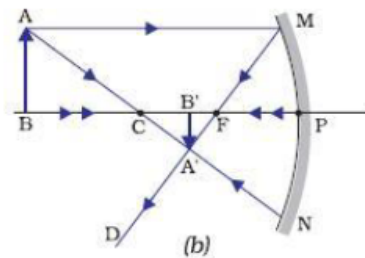
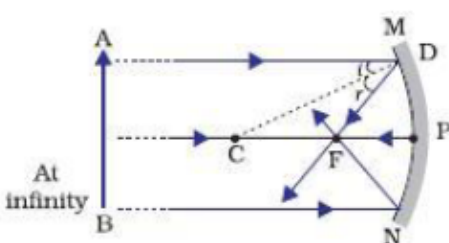
1. Rays parallel to the normal will always diverge away, producing a virtual ray that converges to the virtual focal point.
2. Rays that hit the virtual center of curvature will always reflect back and produce a virtual ray.

Imaging

When drawing the reflected image made by a mirror using a ray diagram, apply all important rays from the tip of the object. The intersection of all rays will be the reflected image.

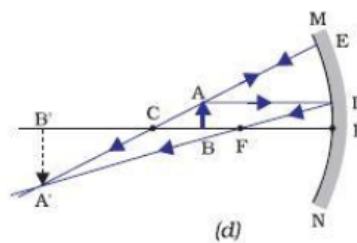
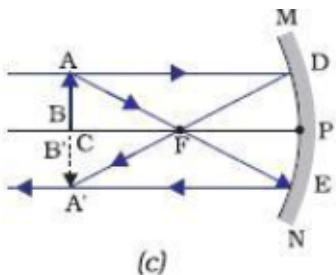
Concave Mirror All Possible Ray Diagrams (and characteristics of image)

Object at infinity (Real Image at Focal) **Object Past C** (Real, inverted, smaller)

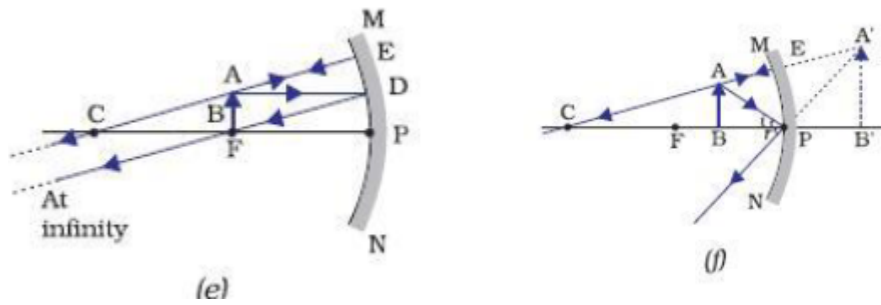


Object at C (Real, Inverted, Same size)

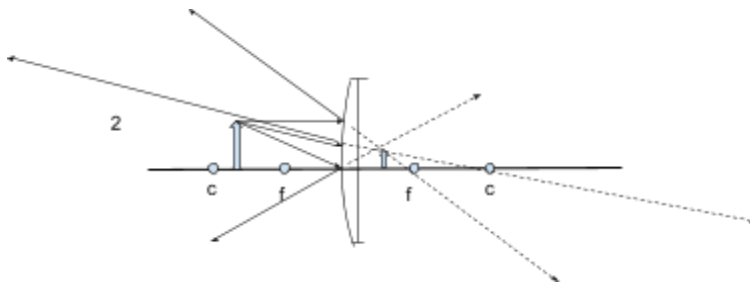
Object Between C and f (Real, inverted, bigger)



Object at f (real, inverted, infinitely big) **Object Between f and mirror** (virtual, erect, larger)



All Ray Diagrams for Convex Mirror (and imaging characteristics)



- All Convex mirrors create smaller, erect and virtual images.

The Mirror Equation and Conventions

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \quad m_{\text{linear}} = \frac{h_i}{h_o} = -\frac{v}{u}$$

f = distance from focal to mirror (+ for concave mirror, - for convex mirror)

v = distance from image to mirror (+ if real image, - if virtual image)

u = distance from object to mirror (+ always)

Lenses

Thin lens: The thin lens is a piece of glass or plastic with regularly curved surfaces which use the phenomenon of refraction to change the direction of light. Thin lenses are the most predictable lens and the only lens observed in option C.

Concave Lens

A concave lens has diverging properties unlike the concave mirror. A concave lens refract rays of light away from the principal axis.

Key Rays

1. All rays parallel to the principal axis refract outwards leaving a virtual ray back to the focal.
2. All rays approaching the focal on the other side are straightened parallel to the principal axis.
3. All rays going through the middle of the lens will go straight through

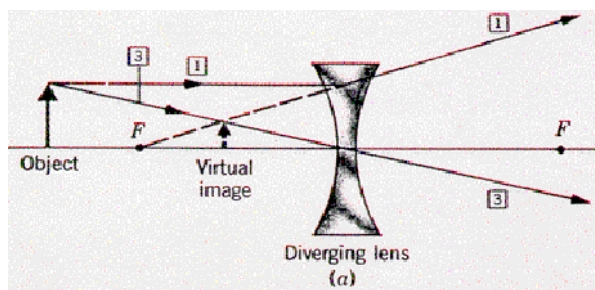
Convex Lens

A convex lens on the other hand possess converging properties and all light rays are refracted towards the principal axis.

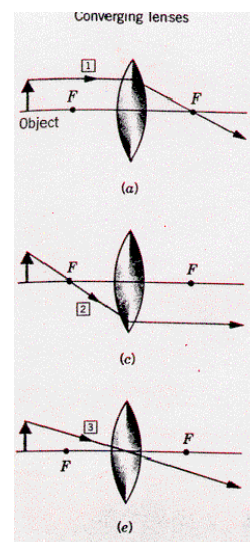
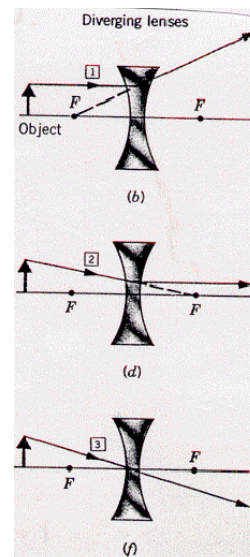
Key Rays

1. Rays parallel to the principal axis will go refract straight into the focal.
2. Rays that cross the focal before hitting the lens will refract parallel to the principal axis.
3. Rays hitting the middle of the lens at the principal axis will go straight through.

Concave Lens Ray Diagrams

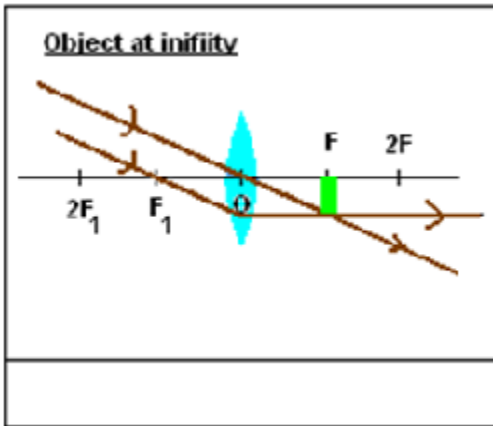


Always smaller, erect and virtual.

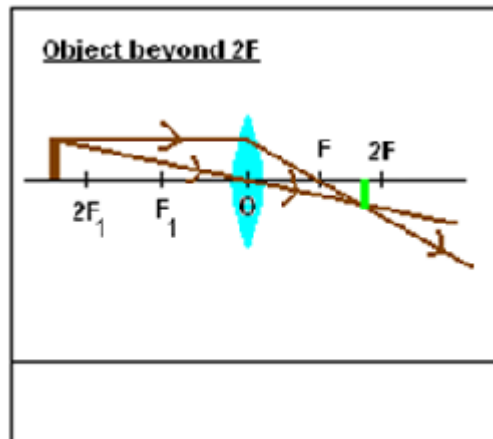


All Convex Ray Diagrams (and image characteristics)

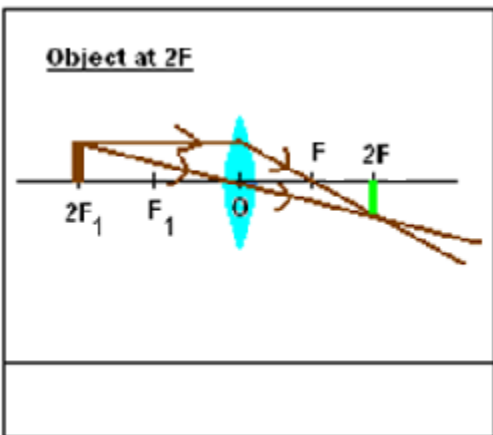
Object at Infinity (real, inverted, at focal)



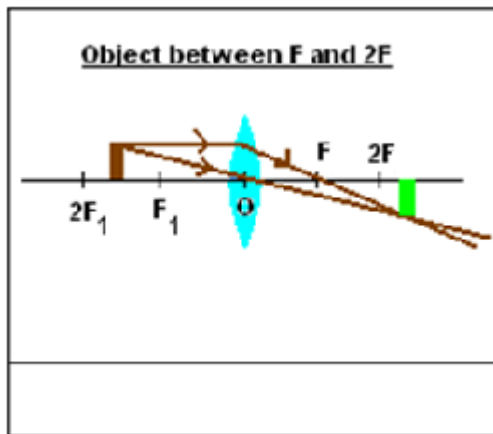
Object Beyond C (real, inverted, smaller)



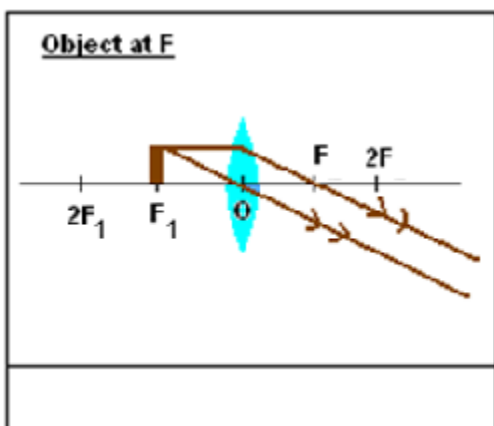
Object at C (real, inverted, same size)



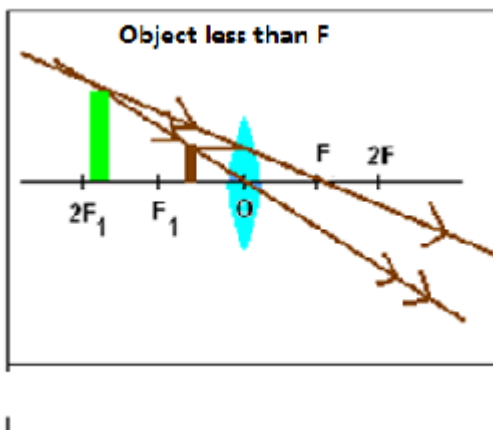
Object Between C and f (real, inverted, larger)



Object at f (real, inverted, at infinity)



Object Between f and lens (virtual, erect, larger)



Note: The mirror equation can be used with lenses just make sure to use correct signs.

Linear and Angular Magnification

Linear magnification is the ratio of image height and object height.

Angular Magnification (Far/Telescope)

$$M_{angular} = \frac{\theta_i}{\theta_o} \quad \theta_i = \frac{h_i}{v} \quad \theta_o = \frac{h_o}{u}$$

Angular Magnification (Close/Microscope)

$$M_{angular} = \frac{\theta_i}{\theta_o} \quad \theta_o = \frac{h_o}{D} \quad \theta_i = \frac{h_i}{v}$$

When an object is far the object can not be moved and therefore the object angle is its height over the distance. But when an object is close the best naked angular size is observed at the near point, D .

Near Point

The near point is the distance an image is most clearly processed by the human eye. The near point is 25 cm. So when using near point sub 25 cm in for the image distance.

Special Formula for Angular Magnification

$$M_{infinity} = \frac{D}{f} \quad M_{near\ point} = 1 + \frac{D}{f}$$

$M_{infinity}$ = angular magnification when object is at focal, image at infinity

$M_{near\ point}$ = angular magnification when image is at near point

D = near point, 25 cm

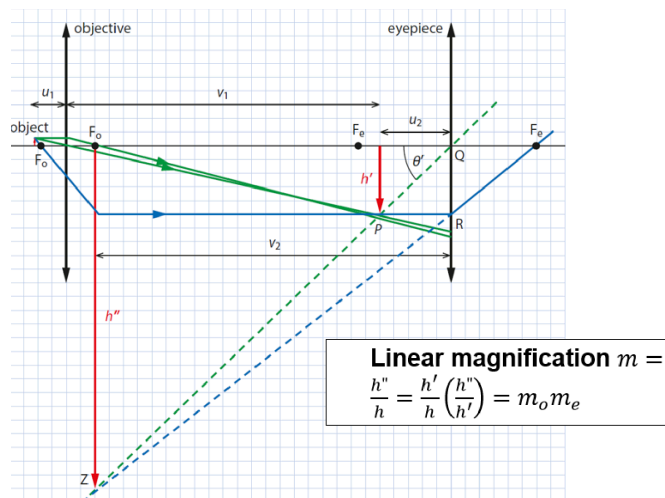
C.2 Imaging Instrumentation

Optical Compound Microscopes

A compound microscope consists of an eyepiece lens and an objective lens. The eyepiece lens adjusts and magnifies the smaller details. While the objective lens is the high magnification lens.

The objective lens should create a real image just in front of the eyepiece focal. The eyepiece lens will then reflect a virtual enlarged image.

$$m_{microscope} = m_o m_e$$



Normal Adjustment

When the final image created by the eyepiece lens is at the near point. The linear magnification and angular magnification are the same.

$$m = M \quad M = m_o m_e \quad M = m_o M_e$$

A compound microscope contains an objective lens of focal length 0.80cm and an eyepiece lens of focal length 5.4cm. The microscope is adjusted to form a final image at the near point of the eye for an object placed 0.92cm in front of the objective lens.

- a. Determine the distance between the two lenses.

$$\frac{1}{f_o} = \frac{1}{v_o} + \frac{1}{u_o} \quad \frac{1}{f_e} = \frac{1}{v_e} + \frac{1}{u_e}$$

$$\text{distance} = v_o + u_e$$

$$f_o = 0.80\text{cm} \quad u_o = 0.92\text{cm} \quad f_e = 5.4\text{cm} \quad v_e = -25\text{cm}$$

$$v_o = \left(\frac{1}{0.80} - \frac{1}{0.92} \right)^{-1} \quad u_e = \left(\frac{1}{5.4} - \frac{1}{-25} \right)^{-1}$$

$$d = 6.13 + 4.44$$

$$d = 10.57 \text{ cm}$$

- b. What is the angular magnification of the image?

Normal Adjustment, Therefore:

$$M = m_o M_e$$

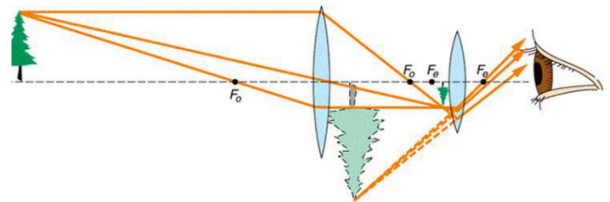
$$M = \left(-\frac{u}{v} \right) \left(\frac{D}{f} + 1 \right)$$

$$M = \left(-\frac{6.13}{0.92} \right) \left(\frac{25}{5.4} + 1 \right)$$

$$M = 37.5$$

Simple optical astronomical refracting telescope

A simple refracting telescope uses two converging lenses with both focal points of the lenses being on top of each other.



Key Formulae (telescopes only)

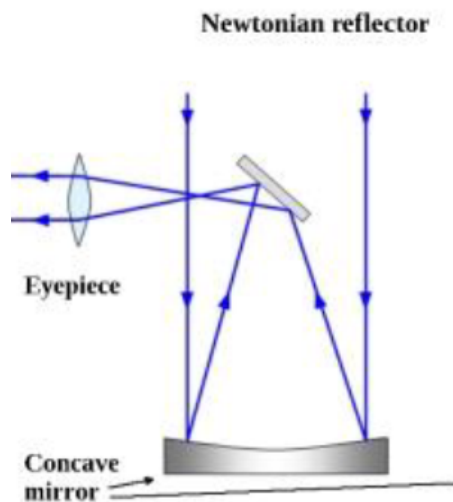
$$l = f_o + f_e \quad M = \frac{f_o}{f_e}$$

Note: All other lens formulas can be used but these only apply to telescopes.

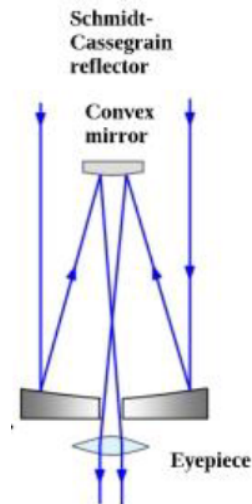
Simple reflecting telescopes

Reflecting telescopes use mirrors to relay the rays from objects very far away into the eye in a condensed enlarged form. There are two types of reflecting telescopes, the Newtonian Reflector and the Cassegrain Reflector.

Newtonian



Cassegrain



Advantages of Reflectors:

- Mirrors are lighter and easier to support
- Reflectors are typically smaller dimensions.
- Mirrors do not suffer chromatic aberration
- Only one side has to be “polished and ground”

Disadvantages of Reflectors

- Easy to get out of alignment
- Tube open to environment
- Supports for secondary mirrors produce a little distortion.
-

Single-dish radio telescopes

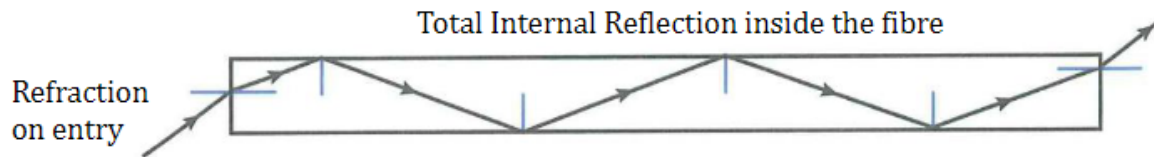
Radio Interferometry Telescopes

Satellite-borne telescopes

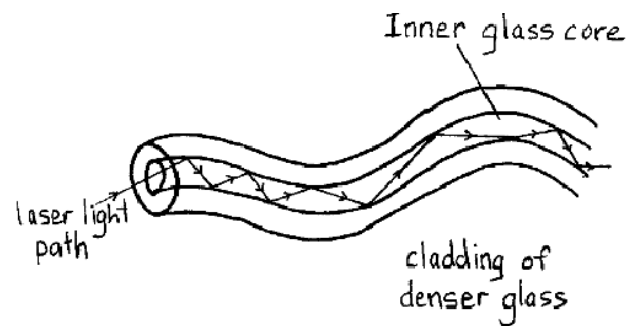
C.3 Fibre Optics

Structure of Optic Fibres

Optical fibers are made to have no light escape and chain instances of total internal inflection, this way the light rays can travel for very long instances while being orientated around obstacles.



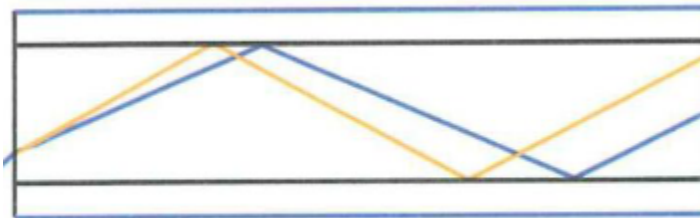
Structure: The interior of an optical fiber is glass and is surrounded by **cladding** which has a significantly higher refractive index. The cladding “traps” the light within the cable.



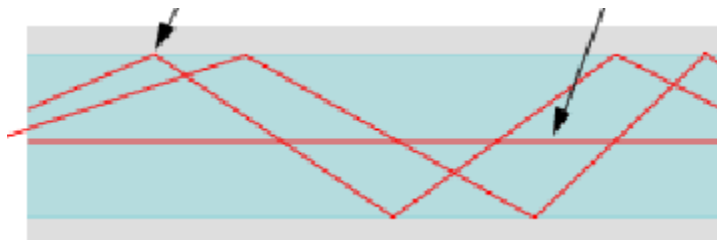
Dispersion

Dispersion is when the signals spread out as they travel along a fibre.

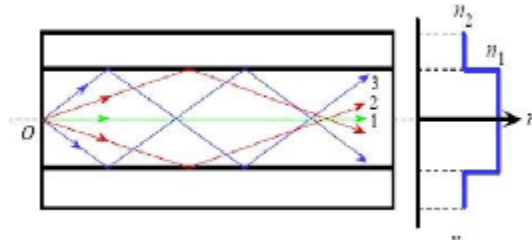
Material Dispersion: As signals travel further and further the thousands of reflections can cause small splits in wavelength. This creates a delay and dispersion in the signal.



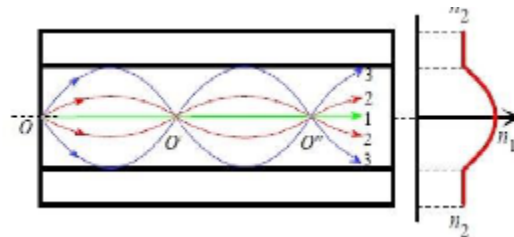
Modal Dispersion or “WaveGuide Dispersion”: Is when multiple signals are sent across an optic fibre and are intentionally sent at different angles so multiple signals can be received.



Step-index Fibre: Step index fibres are used so that rays arrive at different times. Because rays are sent in at different incident angles, each ray has a different path length creating a change in the time it takes for each wave to travel.



Graded Index Fibre: Graded index fibers are cables that allow for multiple waves to be sent and received at the same time. In the glass core of the cable the refractive index varies from highest at the center to the weakest at the ends. This creates many minute refractions until a critical angle is hit, and because waves travel at different speeds for each different path, they will all return at the end.



Attenuation and the Decibel scale

Another problem is that light signals scatter and the power/intensity of these waves decrease. The power loss is called attenuation, measured in decibels (dB)

$$attenuation = 10 \log \frac{I}{I_0}$$

Note: Attenuation/Kilometer is the **specific attenuation** of an optic fiber.